

Lesion-Aware Explainable Hybrid CNN–Transformer Framework for Diabetic Retinopathy Severity Grading

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by

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Under the supervision of

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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Dedicated to my family,
teachers, and mentors.

– *Student Name*

ABSTRACT

Diabetic retinopathy (DR) is a progressive retinal microvascular complication of diabetes. Automated grading remains difficult because clinically important lesions can be subtle, class frequencies are uneven, image acquisition varies across datasets, and neighboring severity grades often share similar appearances. This work develops an explainable hybrid convolutional neural network (CNN)–Transformer for five-class DR severity grading, extends the design with explicit lesion-aware supervision, and implements the completed grading model in a web-based research decision-support prototype, *RetinaInsight*. The grading cohort combines APTOS2019, EyePACS, and Messidor-2 after duplicate removal and manual harmonization to a common five-grade label space, yielding 143,669 fundus images. A single image-level split was created and then locked for every model comparison: 115,241 images for training, 14,227 for validation, and 14,201 for testing. An ImageNet-pretrained EfficientNet-B0 provides hierarchical retinal features, while 49 spatial tokens and four Transformer blocks capture longer-range retinal relationships. Evaluation goes beyond overall classification performance by examining class imbalance, ordinal error, calibration, internal branch behavior, explainability, uncertainty, and controlled image perturbations. On the held-out test set, the completed hybrid model reaches 96.40% accuracy, 0.9630 macro-F1, 0.9790 quadratic weighted kappa, 0.9954 macro ROC-AUC, and 0.9858 macro PR-AUC. Temperature scaling fitted only on the validation set reduces expected calibration error from 0.0610 to 0.0240 and Brier score from 0.0660 to 0.0580 without changing the predicted classes. The lesion-aware extension uses IDRiD masks for microaneurysms, hemorrhages, hard exudates, and soft exudates. It combines an FPN decoder with probability-weighted lesion tokens, lesion-guided cross-attention, lesion-burden features, gated fusion, and an auxiliary ordinal objective. Its evaluation protocol and M0–M4 ablation sequence are fixed in advance, while the corresponding numerical outcomes remain pending. RetinaInsight reproduces the completed 224×224 inference pipeline and presents five-grade probabilities, Grad-CAM++, Transformer attention rollout, uncertainty, and image-quality indicators through a privacy-conscious Flask application. Overall, the thesis links large-scale DR grading with reliability analysis, pathology-aware

representation learning, and a reproducible software translation of the completed model.

Keywords: diabetic retinopathy, fundus imaging, lesion segmentation, EfficientNet, Vision Transformer, multi-task learning, explainable artificial intelligence, ordinal classification, uncertainty, web decision support

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Chapter 1

Introduction

1.1 Background

Diabetes mellitus can produce vascular complications across several organ systems, including the eye. Diabetic retinopathy (DR) is among the most important of these ocular complications because progressive retinal microvascular damage can lead to visual impairment when disease is not identified and managed in time. A large systematic review estimated that 22.27% of people with diabetes had some form of DR in 2020 and projected a substantial increase in the global number of affected adults through 2045 [1]. Broader clinical reviews and incidence studies likewise emphasize the progressive, sight-threatening nature of DR and the importance of timely screening [2, 3, 4]. Taken together, these estimates strengthen the case for scalable screening tools that can help clinicians identify disease while vision-preserving intervention is still possible.

Color fundus photography is widely used in DR screening because it provides a non-invasive view of the retina and captures manifestations such as microaneurysms, hemorrhages, hard exudates, cotton-wool spots, vascular abnormalities, and proliferative changes. The International Clinical Diabetic Retinopathy disease-severity scale organizes disease into ordered categories ranging from no apparent retinopathy to proliferative DR [5]. In automated analysis, these categories are commonly represented as five labels: No DR (grade 0), Mild DR (grade 1), Moderate DR (grade 2), Severe DR (grade 3), and Proliferative DR (grade 4). The ordering is clinically and computationally important because a one-grade error is not equivalent to a distant multi-grade disagreement.

Deep learning has become a central approach to automated retinal image analysis. Large-scale CNN studies demonstrated that automated systems can discriminate referable DR from fundus photographs [6, 7]. Earlier and complementary grading studies established the feasibility of end-to-end CNN staging, automated screening, image-level lesion mining, and multi-stage disease classification [8, 9, 10, 11]. More recent work has moved toward five-class grading, Transformer-based global modeling, lesion-aware representations, calibration, uncertainty, and external validation [12, 13, 14, 15]. Nevertheless, five-grade classification remains difficult because early lesions may occupy only a small portion of the image, neighboring grades share overlapping appearances, and heterogeneous acquisition conditions can introduce non-pathological variation.

The work is organized around three connected contributions. First, it develops and evaluates a large-scale hybrid CNN–Transformer grader using a combined APTOS2019, EyePACS, and Messidor-2 cohort, drawing on three established public screening and grading resources [16, 17, 18]. Second, it extends that grader with explicit IDRiD lesion supervision so that predicted pathological evidence can participate directly in the grading representation. Third, it translates the completed grading model into *RetinaInsight*, a web-based research decision-support prototype that exposes five-class probabilities, uncertainty, image-quality indicators, Grad-CAM++, and Transformer attention rollout.

1.2 Clinical Manifestations Relevant to the Study

1.2.1 Microaneurysms

Microaneurysms (MA) are small focal vascular abnormalities and among the earliest visible manifestations of DR; their automated detection has long been recognized as a technically demanding retinal image-analysis task [19, 20]. Their limited spatial extent makes them challenging for automated systems, especially after aggressive downsampling. The lesion-aware extension therefore operates at a higher resolution than the original grading model and treats MA as an explicit segmentation target.

1.2.2 Hemorrhages

Retinal hemorrhages (HE) vary in morphology and spatial extent, and red-lesion detection studies show the value of combining learned and domain-informed representations for these small structures [21]. Their appearance may overlap with normal dark retinal structures, creating a need for both local texture sensitivity and broader anatomical context. In the lesion-aware pathway, hemorrhage probability maps contribute both pathology-conditioned feature tokens and a normalized burden descriptor.

1.2.3 Hard Exudates

Hard exudates (EX) are bright deposits associated with vascular leakage and have been studied both as direct lesion targets and as evidence for diabetic macular edema assessment [22, 23]. Although visually salient in many images, they can be confused with other bright structures or acquisition artifacts. Explicit EX segmentation therefore offers a more direct pathological target than relying solely on image-level attention.

1.2.4 Soft Exudates

Soft exudates (SE), often referred to as cotton-wool spots, appear as pale retinal lesions and can contribute to more advanced non-proliferative disease patterns. Because lesion pixels occupy a small fraction of the retinal field, segmentation requires imbalance-aware objectives and per-lesion evaluation rather than a single pooled metric.

1.3 Challenges in Automated DR Grading

1.3.1 Small and Subtle Lesions

A grading decision may depend on abnormalities that occupy only a very small part of the retinal field. A classifier that relies predominantly on global appearance may overlook these structures, whereas a purely local model may fail to capture lesion distribution across the retina. The problem is therefore inherently multi-scale:

fine lesion detail must be retained without losing the broader retinal context in which those lesions appear.

1.3.2 Visual Similarity Between Adjacent Grades

Mild and Moderate DR are particularly difficult to separate because they can contain the same lesion types and may differ more in lesion burden or spatial distribution than in entirely distinct visual patterns. Similar ambiguity can arise at other adjacent boundaries. Consequently, this thesis evaluates conventional classification performance together with ordinal error magnitude and quadratic weighted kappa (QWK).

1.3.3 Class Imbalance

The combined grading dataset is strongly imbalanced. Grade 0 contains 68,953 images (47.99%), whereas grade 3 contains 9,914 images (6.90%), producing a majority-to-minority ratio of approximately 6.96:1. Standard cross-entropy can therefore place disproportionate emphasis on well-represented classes. The completed grading workflow uses effective-number class weighting with focal modulation, and the lesion-aware grading pathway preserves the same imbalance-aware principle.

1.3.4 Acquisition and Image-Quality Variation

APTOS2019, EyePACS, and Messidor-2 contain images acquired under different devices and imaging conditions [16, 17, 18]. Brightness, contrast, sharpness, color response, cropping, and field of view can therefore vary. The combined-dataset workflow characterizes these properties before training and later assesses sensitivity to controlled photometric and cropping perturbations.

1.3.5 Local and Global Representation

CNNs provide strong inductive biases for local spatial patterns, whereas self-attention can model longer-range relationships. Major convolutional families such as ResNet, DenseNet, Xception, and ConvNeXt illustrate alternative strategies for depth, connectivity, separable convolution, and modernized ConvNet design [24, 25, 26, 27]. Hierarchical and data-efficient Transformer variants such as Swin,

DeiT, and MaxViT further demonstrate different ways to structure global and local attention [28, 29, 30]. EfficientNet provides a compact hierarchical convolutional representation [31], while Vision Transformer architectures demonstrate the value of token-based global modeling [32, 33]. The original model integrates these mechanisms, and the lesion-aware extension further introduces pathology-specific tokens.

1.3.6 Confidence, Calibration, and Uncertainty

A high softmax probability does not necessarily imply a well-calibrated prediction. Modern neural networks can be overconfident even when discrimination is strong [34]. Predictive uncertainty can be estimated through complementary approaches such as deep ensembles and Bayesian dropout, while uncertainty under distribution shift remains a distinct challenge [35, 36, 37, 38]. Calibration work beyond temperature scaling has also examined Bayesian binning and the calibration behavior of modern architectures [39, 40]. The completed study therefore evaluates expected calibration error (ECE), Brier score, temperature scaling, predictive entropy, and the relationship between uncertainty and ordinal error.

1.3.7 Pathological Grounding of Explanations

Grad-CAM++ and attention rollout are useful for model inspection, but they are not equivalent to expert lesion annotations [41, 42]. The broader XAI literature includes Grad-CAM, integrated gradients, LIME, and SHAP, alongside important critiques showing that saliency and attention should not automatically be interpreted as causal explanation [43, 44, 45, 46, 47, 48, 49, 50]. This distinction motivates the second-stage lesion-aware framework, in which IDRiD masks directly supervise MA, HE, EX, and SE representations rather than treating saliency as lesion ground truth.

1.4 Motivation

The completed grader addresses the local–global representation problem by combining EfficientNet-B0 features with Transformer context, spatial refinement, and a global CNN descriptor. Its evaluation further examines calibration, uncertainty, branch agreement, explainability, and perturbation sensitivity. Even so, image-level

severity labels do not require the network to build an explicit representation of the retinal lesions that underpin clinical grading.

The lesion-aware extension was designed around this limitation. Its central mechanism is not simply the coexistence of segmentation and classification heads. Instead, predicted lesion probability maps are used to perform probability-weighted pooling over learned retinal features, producing one pathology-conditioned token for each lesion type. These lesion tokens condition spatial retinal tokens through cross-attention, after which gated fusion integrates pathology-specific and global context for ordinal DR grading. The methodological hypothesis is therefore:

$$\begin{aligned}
 \text{lesion maps} &\rightarrow \text{lesion-conditioned tokens} \\
 &\rightarrow \text{lesion-guided cross-attention} \\
 &\rightarrow \text{gated pathology-global fusion} \\
 &\rightarrow \text{ordinal DR grading.}
 \end{aligned} \tag{1.1}$$

A third motivation concerns translation beyond the model-development environment. A research model is easier to inspect and reproduce when its inference path can be reconstructed in a controlled interface rather than remaining tied to a training notebook. RetinaInsight therefore reconstructs the completed hybrid model, preserves evaluation-time preprocessing, and provides grade probabilities, complementary XAI maps, uncertainty, and image-quality warnings while maintaining explicit decision-support boundaries. This conservative framing is consistent with broader guidance on responsible medical machine learning and early-stage evaluation of AI-driven decision-support systems [51, 52].

1.5 Problem Statement

Automated five-class DR grading must simultaneously address subtle local pathology, long-range retinal context, class imbalance, ordinal ambiguity, heterogeneous acquisition, confidence reliability, and interpretability. Existing approaches often study these requirements separately: CNNs emphasize local representations; Transformers model global context; lesion-segmentation systems provide pathology localization; and calibration or uncertainty analyses are frequently evaluated after model development rather than integrated into the research design. In addition,

strong image-level grading performance does not by itself establish that the model’s internal representation is grounded in clinically relevant lesions.

The research therefore follows a single progression from large-scale five-grade classification, through reliability-oriented evaluation and explicit lesion-conditioned representation learning, to a reproducible web-based inference prototype.

1.6 Research Aim

The aim of this research is to develop and evaluate an explainable lesion-aware hybrid CNN–Transformer framework for five-class diabetic retinopathy severity grading by integrating large-scale image-level supervision, IDRiD-derived pixel-level lesion supervision, ordinal reasoning, reliability analysis, and a web-based decision-support prototype.

1.7 Research Objectives

1. Develop a hybrid CNN–Transformer model for five-grade DR classification using a combined APTOS2019, EyePACS, and Messidor-2 cohort.
2. Address grading imbalance using class-balanced focal learning and macro-averaged evaluation.
3. Evaluate ordinal consistency using QWK, absolute grade error, within-one-grade accuracy, and major-error rate.
4. Evaluate calibration, predictive uncertainty, local–global agreement, explainability, and controlled perturbation sensitivity.
5. Develop an IDRiD-supervised segmentation pathway for MA, HE, EX, and SE.
6. Introduce probability-weighted lesion tokens, lesion-guided cross-attention, and gated pathology–global fusion for lesion-aware grading.
7. Isolate segmentation supervision, lesion tokens, cross-attention, and ordinal learning through a predefined M0–M4 ablation sequence.

8. Develop *RetinaInsight*, a web-based research decision-support prototype that reproduces the trained grading model’s inference preprocessing and exposes grade probabilities, uncertainty, image-quality information, and complementary XAI outputs.

1.8 Research Questions

1. How effectively does the hybrid CNN–Transformer framework perform five-grade DR classification on the combined grading cohort?
2. How well does the framework handle minority classes and ordinal grading errors?
3. How reliable are its predictions with respect to calibration, uncertainty, branch agreement, explainability, and controlled perturbations?
4. Can IDRiD supervision learn explicit MA, HE, EX, and SE representations suitable for integration with the grading pathway?
5. Can lesion-conditioned tokens and lesion-guided cross-attention provide pathology-aware information beyond the original local–global representation?
6. What are the individual contributions of segmentation supervision, lesion tokens, cross-attention, and ordinal learning under a common 512×512 ablation protocol?
7. Can the completed hybrid grader be translated into a reproducible web-based inference and explanation workflow while preserving preprocessing consistency and explicit clinical-use safeguards?

1.9 Research Contributions

Three contributions define the scope of the work. First, it establishes a large-scale five-class grading framework based on a duplicate-cleaned, manually harmonized combination of APTOS2019, EyePACS, and Messidor-2, with all grading baselines trained under the same locked experimental protocol. Second, it develops a lesion-aware extension whose distinctive contribution is the transformation of

explicit lesion probability maps into pathology-conditioned tokens that directly modulate retinal Transformer features through cross-attention and gated fusion. The extension is evaluated through a predefined M0–M4 sequence rather than by comparing only an auxiliary segmentation head against the original model. Third, it contributes RetinaInsight, a web-based research prototype that reproduces the completed grading inference pipeline and presents predictions, class probabilities, Grad-CAM++, Transformer attention rollout, uncertainty, and image-quality indicators in a privacy-conscious interface.

1.10 Scope and Limitations

The scope is limited to color fundus photographs and five-category DR severity grading. The completed grading cohort combines APTOS2019, EyePACS, and Messidor-2. Duplicate images were removed and source labels were manually mapped to the common five-grade representation before a single image-level train/validation/test split was created and locked. Because the split is image-level, the study does not claim patient-disjoint or eye-disjoint independence. Repeated multi-seed training is also not performed because of computational limitations, so baseline differences are interpreted descriptively rather than through inferential significance testing.

The lesion-aware extension uses IDRiD as the explicit segmentation-supervision source for MA, HE, EX, and SE. Every segmentation sample used by the workflow is represented by four aligned binary lesion channels; an empty channel denotes confirmed absence of that lesion. The lesion-aware protocol operates at 512×512 , whereas the completed original grader operates at 224×224 ; therefore, the M0–M4 sequence is the correct within-protocol ablation for attributing effects to lesion-aware components.

RetinaInsight is a research decision-support prototype rather than a clinical diagnostic system. It does not provide treatment recommendations, patient-record management, authentication, or regulatory validation. The current prototype also reports raw softmax probabilities because the validation-fitted temperature parameter is not bundled with the standalone checkpoint.

1.11 Thesis Organization

Chapter 2 reviews the clinical and technical literature relevant to five-class DR grading, including public datasets, CNN and Transformer methods, lesion segmentation, class imbalance, ordinal learning, explainability, calibration, uncertainty, robustness, and decision-support systems. It closes with a focused five-class SOTA comparison and the research gap addressed here. Chapter 3 presents dataset construction, manual label harmonization, duplicate handling, image-level splitting, data characterization, the original hybrid grader, lesion-aware representation learning, multi-task and ordinal objectives, the experimental protocol, and RetinaInsight. Chapter 4 reports completed grading and software-verification results, presents contextual SOTA analysis, and reserves predefined tables for lesion-aware numerical outcomes. Chapter 5 summarizes the completed findings, states the methodological contribution of the lesion-aware extension without claiming unexecuted benefits, discusses limitations, and identifies future directions.

Chapter 2

Literature Review

2.1 Diabetic Retinopathy and Five-Class Severity Grading

Diabetic retinopathy (DR) is a progressive retinal microvascular complication of diabetes and a major cause of preventable vision impairment. The International Clinical Diabetic Retinopathy scale organizes disease from no apparent DR through mild, moderate, and severe non-proliferative DR to proliferative DR [5]. For machine learning, this ordering matters because the meaning of an error depends on how far the prediction lies from the reference grade. A model that predicts grade 2 instead of grade 1 makes a different type of error from a model that predicts grade 4 for a grade-1 image.

Large-scale epidemiological evidence underscores the need for scalable screening, while reviews of deep-learning grading consistently identify adjacent non-proliferative stages as a persistent source of error [1, 53]. This pattern is consistent with the clinical basis of grading: subtle changes in lesion type, burden, and distribution can separate neighboring classes, whereas No DR and advanced proliferative disease often provide more distinct visual evidence.

2.2 Public Fundus Datasets and Benchmarking

Public retinal datasets are far from interchangeable: they differ in population, camera type, field of view, annotation protocol, and intended task. Foundational public retinal resources for vessels and pathology, including DRIVE and STARE,

helped establish reproducible benchmarking practices in retinal image analysis [54, 55]. EyePACS became a major benchmark for large-scale DR screening and grading and was used in early high-impact deep-learning systems [6, 17]. APTOS2019 later became a common five-class benchmark, while Messidor and Messidor-2 have been widely used to study external and cross-domain behavior [16, 18]. APTOS2019 provides five-grade labels and is widely used for multi-class benchmarking, while Messidor-2 provides a separate European fundus collection often used for external or cross-domain evaluation [56]. IDRiD provides both grading information and expert pixel-level retinal lesion annotations and is therefore especially valuable for connecting image-level severity prediction with explicit pathology [57].

Stage I combines APTOS2019, EyePACS, and Messidor-2 after duplicate removal and manual harmonization to a common five-grade label space. This design increases visual diversity but also creates a need for careful source-aware interpretation because merged datasets can retain device and acquisition cues even when class distributions are balanced across splits. Accordingly, the methodology characterizes brightness, contrast, sharpness, and class distribution and keeps the split fixed across all grading baselines.

2.3 CNN-Based DR Classification

CNNs remain strong retinal-image classifiers because their local connectivity and hierarchical feature extraction are well suited to lesion-sensitive visual patterns. Gulshan et al. demonstrated that deep CNNs could achieve high discrimination for referable DR on large fundus datasets [6], and Ting et al. further demonstrated multi-ethnic validation of retinal deep-learning systems [7]. Later studies extended CNNs to five-stage classification, multi-scale feature extraction, lesion localization, and efficient deployment [58, 59, 60, 61, 62, 63]. Earlier grading and screening studies by Pratt et al., Gargeya and Leng, Quellec et al., and Takahashi et al. illustrate the transition from general CNN screening to explicit severity staging and weakly localized pathological evidence [8, 9, 10, 11]. Reference-standard quality and human-grader variability are also important because label disagreement can materially change the apparent performance of an automated grader [64]. Longitudinal and spectrum-wide systems further demonstrate that retinal deep learning can be extended from static grading toward progression prediction and

joint quality/lesion/grading pipelines [65, 66]. A broader review of fundus-based DR systems also emphasizes that preprocessing, dataset choice, lesion analysis, and clinical translation are tightly coupled design decisions [67].

At the same time, CNN performance is often sensitive to preprocessing choices, image resolution, and class distribution. Early-stage lesions occupy few pixels, while higher-grade patterns may require information distributed across the retina. CNNs can increase receptive field through depth, but this does not provide explicit token-to-token long-range interaction. These considerations motivate hybrid designs in which convolution remains responsible for local retinal structure while an attention module models broader context.

2.4 Vision Transformers for DR Grading

Vision Transformers (ViTs) extend self-attention to visual representations, allowing information to be exchanged globally among image tokens [32, 33]. In DR grading, this property is attractive because lesion distributions and relationships across distant retinal regions can contribute to severity assessment.

Gu et al. combined a Vision Transformer with residual class-specific attention for five-stage DR classification and reported strong competitive results on DDR while identifying the persistent difficulty of mild DR [12]. Huang et al. introduced saliency-guided self-supervised Transformer pretraining to improve DR grading with domain-specific prior information [14]. Yang et al. explored masked-autoencoder pretraining for large retinal images and used EyePACS, AP-TOS, and Messidor-2 as complementary sources for representation learning and evaluation [56]. Taken together, these studies show the value of global token interaction while also highlighting two unresolved issues: preserving small lesion detail and maintaining performance across datasets. CoT-XNet explicitly combines contextual Transformer and Xception representations, while CrossFiT models complementary information across two fundus fields [68, 69]. Domain-generalization studies further show that calibration, acquisition artifacts, class imbalance, and unseen-domain shifts remain central concerns even with Transformer backbones [70, 71]. Self-supervised and contrastive representation learning provide additional routes to improve data efficiency and transferable visual features [72, 73, 74]. Recent foundation-model work such as RETFound further demonstrates the value of

retina-specific self-supervised representation learning for generalizable ocular tasks [75].

2.5 Hybrid CNN–Transformer Grading

Hybrid CNN–Transformer designs are a natural response to this trade-off, retaining convolutional sensitivity to local lesions while using attention to model broader retinal context. Related attention designs such as squeeze-and-excitation and CBAM show how channel- and spatial-attention mechanisms can modulate convolutional representations before or alongside Transformer reasoning [76, 77]. CRA-Net uses category-relation and lesion-relation attention under Transformer guidance to address class similarity and lesion interactions [13]. ADCNet similarly explores lesion-aware information propagation across cascaded attention stages without requiring dense lesion masks [78]. D-TNet combines DenseNet and Transformer representations for five-grade classification [79]. FusionNet uses parallel CNN and ViT branches for five-level severity assessment and reports strong performance on mixed APTOS2019 and Messidor-2 data with external testing [80]. EffTNet further introduces attention-augmented feature fusion and pair-aware contrastive learning on a harmonized APTOS2019–EyePACS–Messidor-2 training set [15].

The literature therefore supports a local–global design, although the way the two representations interact remains important. Concatenating or late-fusing independent CNN and Transformer embeddings does not explicitly establish which pathological cues influence global token reasoning. The lesion-aware extension proposed here targets this narrower gap by converting supervised lesion probability maps into pathology-conditioned tokens that directly participate in cross-attention with retinal spatial tokens.

2.6 Retinal Lesion Segmentation

Pixel-level lesion segmentation introduces a more explicit form of pathological supervision than image-level grading alone. U-Net established the encoder–decoder paradigm for biomedical segmentation [81], and feature pyramid networks provide a general strategy for fusing multi-scale features [82]. In DR, segmentation is

particularly imbalanced because microaneurysms and other lesions can occupy a very small fraction of image pixels.

IDRiD provides expert annotations for microaneurysms, hemorrhages, hard exudates, and soft exudates [57]. Recent work has evaluated recurrent U-Net variants, lesion-aware optimization, and joint lesion-localization systems on IDRiD and DDR [83, 84, 85]. Such systems demonstrate that lesion localization can be learned from public retinal datasets, but segmentation accuracy alone does not determine whether the resulting lesion representation improves five-grade grading. The DDR benchmark is also important because it combines image-level grading with a lesion-annotated subset, enabling multi-task research at larger scale [86].

The segmentation objective also requires imbalance-aware losses. Dice-based overlap objectives reduce background dominance, while Tversky loss allows false-positive and false-negative penalties to differ. Focal Tversky loss further concentrates optimization on difficult examples [87]. These principles motivate the Dice plus Focal-Tversky objective used in the lesion-aware extension.

2.7 Lesion-Aware and Multi-Task DR Learning

Multi-task learning can connect lesion segmentation and severity grading at different levels of integration. A weak form shares an encoder but keeps segmentation and classification largely independent. A stronger form allows predicted lesion features to modulate the grading pathway. Recent joint multi-task work has used cross-task attention to condition grading representations on lesion features and has shown that removing cross-task attention can reduce grading performance [88]. Other recent studies combine Transformer-based lesion classification and grading to model lesion-sensitive severity information [89]. Other frameworks explicitly quantify lesion information for interpretable stage prediction [90].

The present lesion-aware framework differs from a lesion-count-only pipeline. Predicted lesion maps are used to perform probability-weighted pooling over a learned intermediate feature map, preserving texture and contextual information in four lesion-conditioned tokens. Retinal spatial tokens then query these lesion tokens through multi-head cross-attention before a Transformer encoder and gated fusion. This creates a direct differentiable path from explicit lesion supervision to the final grading representation.

2.8 Class Imbalance and Minority-Grade Learning

Five-class DR datasets are frequently imbalanced, with No DR and Moderate DR often more common than Severe DR. Focal loss reduces the contribution of easy examples and emphasizes harder samples [91]. Class-balanced loss based on the effective number of samples provides a smoother alternative to direct inverse-frequency weighting [92]. These methods are useful when the objective is not only high overall accuracy but also stable macro-F1 and sensitivity across minority classes.

The combined grading cohort in this thesis has a global majority-to-minority ratio of approximately 6.96:1. The final grading objective therefore combines effective-number class weights with focal modulation and label smoothing. The same weighting principle is used for all within-study baseline models to preserve protocol fairness.

2.9 Ordinal Learning for DR Severity

DR grades are categories with an inherent order. Conventional softmax classification treats all incorrect labels as distinct but does not explicitly encode severity distance. QWK addresses this at evaluation time by assigning larger penalties to distant disagreements. Ordinal learning can also be introduced during training through cumulative threshold targets such as $y > 0$, $y > 1$, $y > 2$, and $y > 3$.

The lesion-aware M4 configuration adds an auxiliary ordinal binary-cross-entropy objective to the five-class grading head. This design preserves calibrated multi-class probabilities while regularizing the shared representation with ordered severity information. The ordinal contribution is isolated in the M0–M4 ablation rather than being assumed beneficial.

2.10 Explainable Artificial Intelligence in DR

Explainability is widely used in retinal AI to inspect whether model sensitivity overlaps plausible pathological regions. Grad-CAM++ generates class-sensitive localization from convolutional activations and gradients [41], while Transformer attention rollout aggregates attention relationships across layers [42]. Recent DR

studies have used saliency, Grad-CAM, LIME, and attention mechanisms to visualize retinal evidence [93, 80, 59].

However, post-hoc maps should not be equated with lesion segmentation or causal explanation. Saliency can be spatially coarse, attention can emphasize acquisition artifacts, and visually plausible maps can accompany incorrect predictions. The thesis therefore treats XAI as model-inspection evidence and introduces IDRiD segmentation as a stronger form of direct pathological supervision.

2.11 Calibration and Predictive Uncertainty

Discrimination and calibration describe different properties of a classifier. A classifier can have high accuracy while assigning probabilities that do not correspond to empirical correctness. Temperature scaling is a simple post-hoc calibration method that optimizes a scalar temperature on validation logits while preserving class ordering [34]. In medical AI, calibration is particularly relevant when confidence may be used for referral, triage, or human review.

The completed grading study evaluates ECE and multiclass Brier score before and after validation-fitted temperature scaling. It also computes normalized predictive entropy for each test image and examines accuracy, mean grade error, and major-error rate across entropy quintiles. Recent calibration-focused retinal benchmarks similarly emphasize that strong representation quality should be evaluated together with probability reliability and cross-domain behavior [94]. Recent five-stage systems also increasingly report explainability and preprocessing sensitivity, although calibration remains less common than accuracy or AUC reporting [59, 15].

2.12 Robustness and Domain Generalization

Variation in camera, field of view, population, illumination, and image quality makes cross-dataset retinal evaluation particularly demanding. Cross-domain studies often show meaningful performance degradation when models are moved from one dataset to another [14, 15, 94]. This motivates explicit distinction between within-cohort performance and external generalization.

The completed study characterizes acquisition variation and applies controlled brightness, contrast, blur, color, and cropping perturbations as sensitivity analysis.

Related work on group-enhanced Transformers for retinal severity classification further illustrates the continued interest in modeling subtle stage-level lesion variation with attention mechanisms [95]. These tests are not substitutes for external validation. They are diagnostic experiments intended to identify which non-pathological image changes most affect confidence and adjacent-grade probability structure.

2.13 Web-Based Decision Support and Translational Prototypes

Translating a trained model into a usable application involves more than exporting its weights. Inference preprocessing must match the evaluation pipeline; file handling and error states must be controlled; explanation outputs must be clearly bounded; and probability displays must not be framed as clinical certainty. Regulatory and clinical literature on autonomous retinal AI emphasizes the difference between retrospective algorithm performance and validated clinical deployment [96, 7].

RetinaInsight is positioned at the research-prototype level. It reconstructs the completed Hybrid CNN–Transformer architecture, performs the same 224×224 evaluation-time preprocessing, produces five-class probabilities, Grad-CAM++, attention rollout, uncertainty, and image-quality indicators, and does not persist uploaded images to disk. It is intentionally not presented as a diagnostic device, and the current standalone implementation does not apply the separately fitted temperature scalar.

The reliability literature also cautions against treating confidence, saliency, and in-distribution accuracy as interchangeable notions. Deep ensembles, Bayesian approximations, confidence-based out-of-distribution baselines, and uncertainty-under-shift benchmarks address different aspects of predictive reliability [35, 36, 37, 97, 38, 40]. Robustness benchmarks and in-the-wild distribution-shift studies show that strong development-set performance can degrade under corruptions or population changes [98, 99, 100]. In medical AI, interpretability and deployment guidance additionally emphasize that explanations and user-facing interfaces require validation as socio-technical decision-support components rather than as isolated visualization features [101, 51, 52].

2.14 State-of-the-Art Context for Five-Class DR Grading

Table 2.1 summarizes representative recent five-class DR grading studies. The values are reported only for contextual comparison because datasets, splits, pre-processing, training schedules, and averaging conventions differ. They must not be interpreted as a controlled ranking against the same-protocol experiments in Chapter 4.

Table 2.1: Representative recent literature for five-class DR severity grading. Published values are contextual and are not directly comparable across different datasets and protocols.

Study	Year	Dataset(s)	Main approach	Accuracy	F1	Kappa/QWK	XAI / validation
Gu et al. [12]	2023	DDR / IDRiD	ViT with residual class-specific attention	0.8235 ^a	–	–	Attention; multi-dataset
Islam et al. [58]	2022	APTOS2019 / Messidor-2	Xception + supervised contrastive learning	84.36%	Reported	–	t-SNE / cross-dataset
CRA-Net [13]	2024	DDR / AP-TOS2019	Category-relation and lesion-relation attention with Transformer guidance	Competitive	Competitive	Reported	Interpretable attention
HFF-Net [60]	2024	Public DR grading cohort	Hierarchical multi-scale CNN feature fusion	73.77%	–	–	Efficient CNN
D-TNet [79]	2025	APTOS2019 / Messidor-2	DenseNet–Transformer late fusion	97% / 86%	0.94 / 0.79	$\kappa = 0.93$ (APTOS)	Cross-dataset drop
TOViT [102]	2025	EyePACS / Messidor-2 / AP-TOS2019	Task-optimized Vision Transformer	up to 99%	>0.93	Reported	Resource-constrained deployment
FusionNet [80]	2026	APTOS2019 + Messidor-2; external sets	Parallel GoogLeNet and ViT fusion	98.85%	97.62% weighted	–	External testing
EffTNet [15]	2026	APTOS2019 + EyePACS + Messidor-2; external sets	EfficientNet-B3a + ViT-B16 with attention-augmented fusion	98.77%	96.99%	0.947 QWK	Grad-CAM; external validation

^aThe published Gu et al. ablation reports 0.8235 average accuracy for the ViT+CSRA configuration on DDR; its per-class accuracy values are not equivalent to overall sample accuracy. Dashes indicate metrics that were not consistently reported in the source format used for this contextual table.

Two broader patterns emerge from the table. First, recent five-class grading increasingly uses attention, Transformers, feature fusion, and multi-dataset evaluation. Second, headline accuracy varies substantially with the underlying benchmark and protocol. For this reason, the thesis maintains a strict separation between published SOTA context and the same-protocol baseline comparison in Chapter 4.

2.15 Research Gap

Although the literature has advanced five-class grading, lesion localization, Transformer modeling, calibration, uncertainty, and deployment, these capabilities are still commonly evaluated in isolation. Hybrid CNN–Transformer models typically fuse local and global visual embeddings, while lesion-aware systems may use segmentation as an auxiliary task or reduce pathology to lesion counts. Post-hoc explanations provide interpretive maps but do not guarantee pixel-level pathological grounding. Calibration and uncertainty are also inconsistently reported, and deployment prototypes may not preserve the same preprocessing or reliability safeguards used during model evaluation.

The gap addressed by this thesis is therefore a connected research design that combines: (i) large-scale five-grade grading from a duplicate-cleaned and manually harmonized APTOS2019–EyePACS–Messidor-2 cohort; (ii) local CNN and global Transformer modeling under a common baseline protocol; (iii) reliability analysis through calibration, uncertainty, branch agreement, XAI, and perturbation sensitivity; (iv) explicit IDRiD lesion supervision transformed into probability-weighted pathology tokens; (v) lesion-to-retinal cross-attention and gated fusion for ordinal grading; and (vi) a reproducible web-based research prototype that exposes the completed model’s predictions and explanations without overstating clinical readiness.

2.16 Chapter Summary

The literature reviewed in this chapter spans clinical five-class DR grading, public retinal datasets, CNN and Transformer methods, hybrid grading, lesion segmentation, multi-task lesion-aware learning, class imbalance, ordinal classification, explainability, calibration, uncertainty, robustness, and translational decision-support systems. Recent five-class SOTA studies demonstrate strong performance but also substantial protocol dependence, reinforcing the need to distinguish contextual literature comparisons from controlled within-study baselines. These findings motivate the methodology presented in Chapter 3.

Chapter 3

Proposed Method

3.1 Overall Research Framework

The research framework is organized into three connected components. Stage I covers the completed large-scale five-grade grading study. It uses a duplicate-cleaned and manually harmonized combination of APTOS2019, EyePACS, and Messidor-2 to train an EfficientNet-B0–Transformer classifier and evaluates discrimination, ordinal error, calibration, branch agreement, uncertainty, explainability, and controlled perturbation sensitivity. Stage II is the lesion-aware extension. It retains EfficientNet-B0 as the shared encoder, adds an IDRiD-supervised four-channel lesion decoder, converts predicted lesion probability maps into pathology-conditioned tokens, introduces lesion-guided cross-attention and gated fusion, and adds an auxiliary ordinal objective. The third component is *RetinaInsight*, a web-based research decision-support prototype that reconstructs the completed Stage-I model and exposes five-grade prediction, class probabilities, complementary XAI maps, uncertainty, and image-quality indicators.

3.2 Dataset Construction and Characterization

3.2.1 Source Datasets and Combined Grading Cohort

The grading cohort was assembled from three public color-fundus resources: APTOS2019, EyePACS, and Messidor-2. Because the three collections differ in acquisition conditions and source metadata, they were treated as heterogeneous sources rather than as visually interchangeable datasets. Before the final cohort was created,

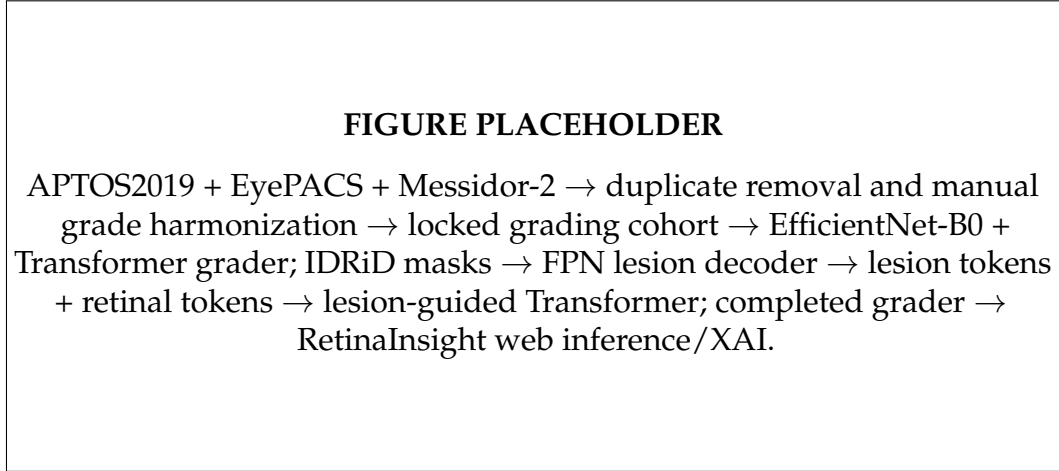


Figure 3.1: Overall research framework linking large-scale five-grade grading, IDRiD-supervised lesion-aware representation learning, and the RetinaInsight research prototype.

duplicate images were removed and source labels were manually mapped to the common five-grade DR severity representation used throughout the thesis:

$$\mathcal{Y} = \{0, 1, 2, 3, 4\} = \{\text{No DR, Mild, Moderate, Severe, Proliferative DR}\}. \quad (3.1)$$

The final combined dataset contains 143,669 images. Source-wise retained counts are not required for model execution because the experiment uses the final consolidated manifest; however, source identities and the manual harmonization process are stated explicitly for provenance transparency.

Table 3.1: Source datasets used to construct the combined grading cohort.

Source	Role	Image type	Label treatment	Duplicate handling
APTOS2019	Grading	Color fundus	Manual mapping to grades 0–4	Removed before final cohort
EyePACS	Grading	Color fundus	Manual mapping to grades 0–4	Removed before final cohort
Messidor-2	Grading	Color fundus	Manual mapping to grades 0–4	Removed before final cohort
IDRiD	Lesion supervision	Color fundus + masks	Four aligned lesion channels	Audited against grading cohort

3.2.2 Image-Level Locked Split

After harmonization, one image-level train/validation/test partition was created and kept fixed for every completed grading experiment and same-protocol baseline. The split is not described as patient-disjoint or eye-disjoint because patient identity

Table 3.2: Combined grading dataset distribution by split and DR grade.

Split	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4	Total
Training	55,162	18,470	24,198	7,936	9,475	115,241
Validation	6,895	1,840	3,024	1,000	1,468	14,227
Test	6,896	1,862	2,999	978	1,466	14,201
Total	68,953	22,172	30,221	9,914	12,409	143,669

was not used to construct it. The final partition contains 115,241 training, 14,227 validation, and 14,201 test images.

Across the full cohort, grades 0–4 account for 47.99%, 15.43%, 21.04%, 6.90%, and 8.64%, respectively. The global majority-to-minority ratio is approximately 6.96:1.

3.2.3 Class Imbalance, Gini Index, and Concentration

Class imbalance was examined for the full cohort as well as for each of the three fixed splits. Table 3.3 reports the imbalance ratio (IR) and Gini summary used during data analysis.

Table 3.3: Class-imbalance summary of the combined grading cohort.

Split	Total	IR	Gini	Majority count	Majority class	Minority class
Global	143,669	6.96	0.6894	68,953	Grade 0	Grade 3
Training	115,241	6.95	0.6896	55,162	Grade 0	Grade 3
Validation	14,227	6.89	0.6876	6,895	Grade 0	Grade 3
Test	14,201	7.05	0.6870	6,896	Grade 0	Grade 3

A Lorenz curve and a cumulative class-distribution plot provide a second view of this concentration. Grades 0–2 account for the majority of the cohort, while Severe and Proliferative DR form a smaller proportion. These observations motivate class-balanced optimization and macro-averaged reporting rather than accuracy alone.

3.2.4 Class-Distribution Consistency Across Splits

Kullback–Leibler (KL) divergence was used to compare class proportions across the fixed splits. The train–validation and train–test divergences are approximately 0.0058 and 0.0054, respectively, while validation–test divergence is close to zero.

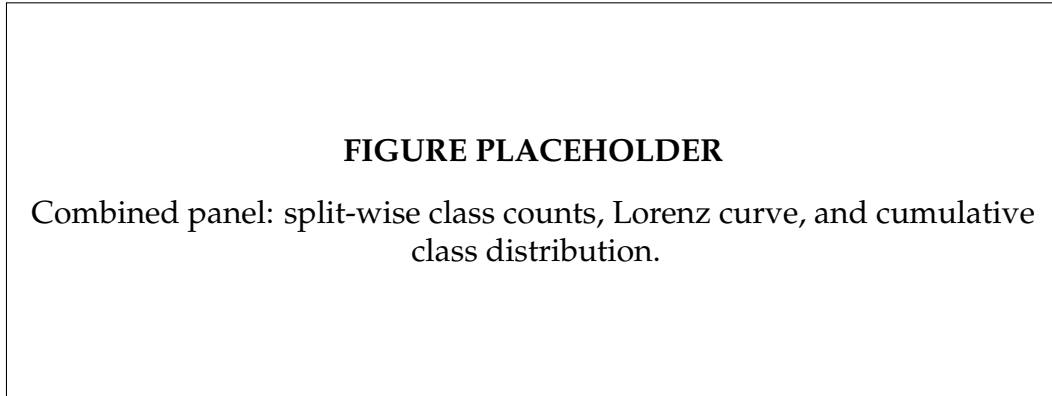


Figure 3.2: Distribution and concentration of the five DR grades in the combined cohort.

The small divergences show that the three partitions have closely matched label proportions. This analysis does not establish image or patient independence; it only shows that the class-proportion distributions are closely aligned.

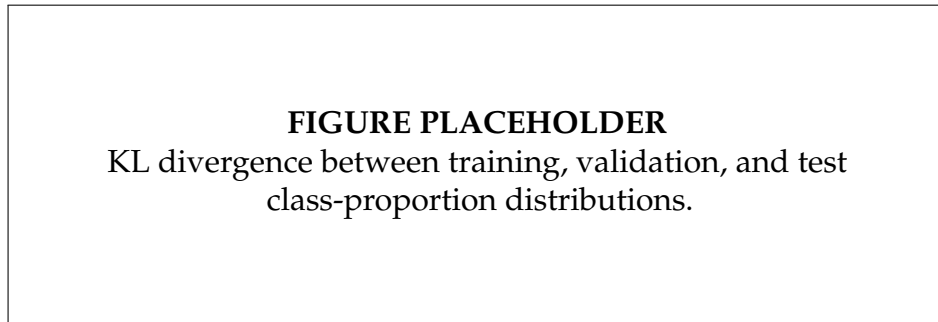


Figure 3.3: Class-distribution similarity across the locked grading splits.

3.2.5 Training-Set Image-Quality Characterization

Brightness, contrast, and sharpness were measured to characterize non-pathological variation in image acquisition. Brightness is computed from grayscale intensity, contrast from grayscale intensity dispersion, and sharpness from the variance of the Laplacian operator. They were used to describe the cohort, not to remove images retrospectively.

The full distributions show broad within-class variation and outliers, indicating that image quality is not uniquely associated with one severity class. This supports

Table 3.4: Mean training-set image-quality descriptors by DR grade.

DR grade	Mean brightness	Mean contrast	Mean sharpness
0	101.15	25.80	47.54
1	109.83	26.66	65.24
2	101.26	25.30	43.71
3	107.87	24.36	55.35
4	110.77	26.37	53.76

controlled augmentation and later perturbation analysis while avoiding a causal claim that quality variation itself produces model error.

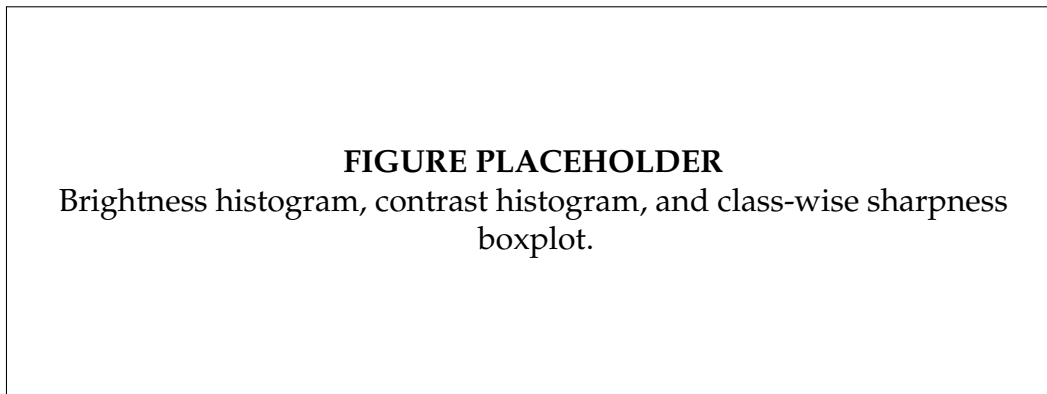


Figure 3.4: Image-quality characterization of the combined grading cohort.

3.2.6 Raw-Feature Separability and t-SNE

For exploratory visualization, t-SNE was applied to 1,500 randomly sampled fundus images after resizing them to 64×64 and flattening the raw pixels. The projection shows substantial overlap among DR grades rather than clean class-separated clusters. The projection is interpreted only as an exploratory view because t-SNE depends on sampling and hyperparameter choices and is not a formal test of class separability. Its main implication is that raw intensity space is insufficient to explain five-grade separation and motivates learned nonlinear local-global features.

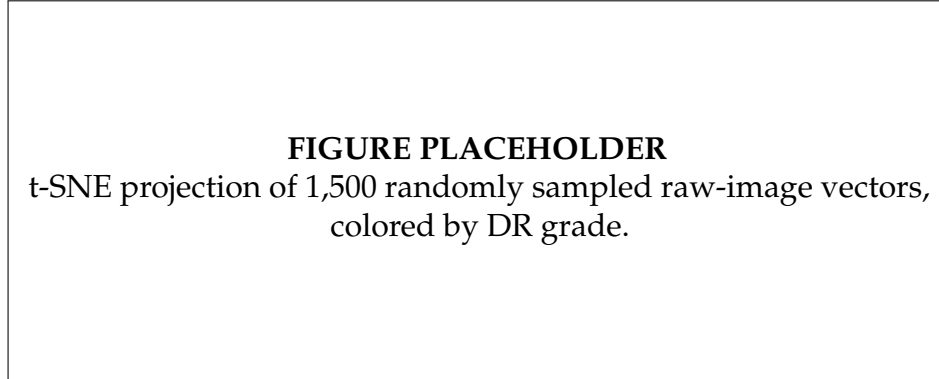


Figure 3.5: Exploratory t-SNE visualization of raw-image feature overlap across DR grades.

3.2.7 IDRiD Lesion-Segmentation Dataset

IDRiD serves exclusively as the source of explicit lesion supervision for the lesion-aware workflow [57]. The official segmentation release is separated into 54 development images and 27 held-out test images. Four lesion channels are modeled:

$$\mathcal{C}_L = \{\text{MA}, \text{HE}, \text{EX}, \text{SE}\}. \quad (3.2)$$

Every segmentation image used by the workflow is represented by four aligned binary masks. When a lesion is absent, the corresponding channel is an empty binary mask; it is not treated as missing annotation. The 27 official test images are excluded from cross-validation, checkpoint selection, threshold development, and early stopping.

Although the wider IDRiD resource includes DR grades, the current segmentation loader does not use those grades for training. This keeps the supervision roles explicit: the combined grading cohort supplies image-level severity labels, whereas IDRiD supplies lesion masks.

3.2.8 Duplicate and Leakage Controls

Duplicate removal is performed while constructing the APTOS2019–EyePACS–Messidor-2 grading cohort. The lesion-aware integration additionally provides exact SHA-256 and perceptual-hash screening to detect possible overlap between IDRiD images and the locked grading manifest. If overlap is detected, the original grading split is authoritative and no image is permitted to cross train/validation/test

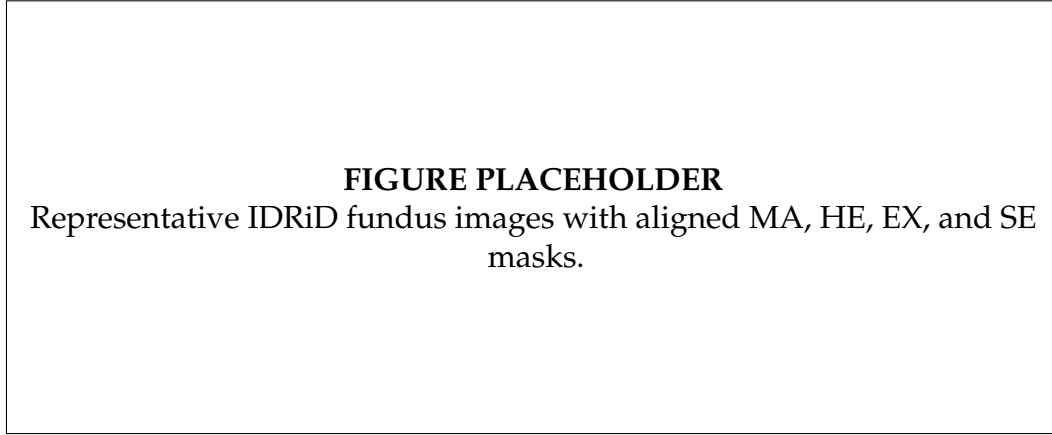


Figure 3.6: Representative lesion supervision from IDRiD.

boundaries through lesion supervision. These controls address image-level duplication; they do not imply patient-level independence.

3.3 Image Preprocessing and Data Augmentation

3.3.1 224-by-224 Grading Protocol

The completed Stage-I grader operates on 224×224 RGB images. A retinal-field crop removes dark borders by locating the visible fundus region. Training augmentation includes random resized cropping, horizontal flipping, rotation within approximately $\pm 15^\circ$, controlled color jitter, and low-probability mild Gaussian blur. Images are then converted to tensors and normalized using ImageNet statistics. Validation and test preprocessing are deterministic. Contrast-limited adaptive histogram equalization is available as an optional configuration but is disabled in the final executed pipeline.

3.3.2 512-by-512 Lesion-Aware Protocol

The lesion-aware extension operates at 512×512 to preserve small lesion structures. Image and all four masks are cropped using the same retinal field, square-padded, and resized; bilinear interpolation is used for RGB images and nearest-neighbor interpolation for masks. During training, image and masks receive identical horizontal flips and rotations. Photometric changes affect only the image. This higher-

resolution protocol is distinct from the completed 224×224 Stage-I experiment and is used consistently across M0–M4.

3.4 Original Hybrid CNN–Transformer Grader

3.4.1 EfficientNet-B0 Feature Extraction

The original model uses an ImageNet-pretrained EfficientNet-B0 backbone [31]. For input

$$X \in \mathbb{R}^{3 \times 224 \times 224}, \quad (3.3)$$

the encoder returns an intermediate feature map F_p and a final convolutional map F_g :

$$(F_p, F_g) = E_\theta(X). \quad (3.4)$$

The intermediate tap retains spatial detail for token construction, while F_g provides a high-level global CNN descriptor.

3.4.2 Spatial Tokenization and Transformer Modeling

A learnable convolution projects F_p to embedding dimension $d = 256$ using kernel size 7, stride 4, and padding 3. The resulting 7×7 grid is flattened to 49 spatial tokens:

$$T_p \in \mathbb{R}^{49 \times 256}. \quad (3.5)$$

A class token and learnable positional embedding are then added. Four pre-normalized Transformer blocks use eight attention heads and a feed-forward expansion ratio of 4.0. The resulting class representation is $z_T \in \mathbb{R}^{256}$.

3.4.3 Spatial Refinement and CNN Descriptor

The 49 non-class tokens are rearranged into a 7×7 feature map and refined by a residual convolution–batch-normalization–GELU block. Global averaging yields $z_S \in \mathbb{R}^{256}$. The final EfficientNet feature map is globally averaged and projected to obtain a second 256-dimensional representation z_C .

3.4.4 Local–Global Fusion

The final feature vector concatenates the Transformer class representation, spatially refined representation, and global CNN descriptor:

$$z_F = [z_T; z_S; z_C] \in \mathbb{R}^{768}. \quad (3.6)$$

A layer-normalized MLP with dropout [103] and a five-unit output layer produces logits for grades 0–4.

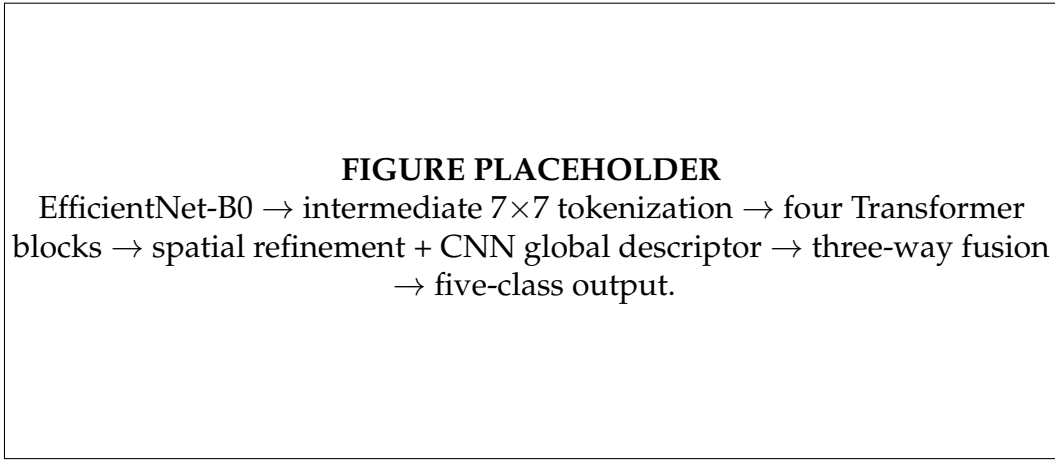


Figure 3.7: Architecture of the completed hybrid CNN–Transformer grading model.

3.5 Lesion-Aware Representation Learning

3.5.1 Multi-Scale EfficientNet Encoder and FPN Decoder

The lesion-aware model preserves EfficientNet-B0 for continuity with Stage I but extracts four multi-scale feature maps from encoder stages with expected channels 24, 40, 112, and 1280. An FPN-style decoder [82] projects the selected features to a common width of 128, performs top-down interpolation and lateral fusion, and uses convolution–batch-normalization refinement blocks [104] before predicting four lesion logits at the input resolution:

$$S = \{S_{MA}, S_{HE}, S_{EX}, S_{SE}\} \in \mathbb{R}^{4 \times 512 \times 512}. \quad (3.7)$$

Independent sigmoid functions produce four lesion-probability maps $P_k = \sigma(S_k)$.

3.5.2 Lesion-Conditioned Feature Pooling

An intermediate encoder feature is projected to a 256-channel lesion feature map F_L . For lesion k , its probability map is resized to the spatial size of F_L , and probability-weighted pooling computes

$$z_k = \frac{\sum_p P_k(p) F_L(p)}{\sum_p P_k(p) + \epsilon}. \quad (3.8)$$

A learnable lesion-class embedding is added to each pooled vector, producing four pathology-conditioned tokens

$$T_L = [z_{MA}, z_{HE}, z_{EX}, z_{SE}] \in \mathbb{R}^{4 \times 256}. \quad (3.9)$$

This mechanism is the central distinction from a conventional auxiliary segmentation head because the lesion probabilities directly construct features used for grading.

3.5.3 Lesion-Burden Representation

For each lesion channel, normalized mean probability is calculated as

$$b_k = \frac{1}{HW} \sum_{u=1}^H \sum_{v=1}^W P_k(u, v). \quad (3.10)$$

The vector $b = [b_{MA}, b_{HE}, b_{EX}, b_{SE}]$ is projected through a small MLP. These values are differentiable summary features and are not interpreted as clinically validated lesion counts.

3.5.4 Retinal Spatial Tokens and Lesion-Guided Cross-Attention

The lesion-aware model projects the intermediate 112-channel encoder feature to 256 channels and uses adaptive average pooling to preserve a compact 7×7 grid:

$$T_S \in \mathbb{R}^{49 \times 256}. \quad (3.11)$$

The retinal spatial tokens act as queries while lesion tokens act as keys and values:

$$Q = T_S W_Q, \quad (3.12)$$

$$K = T_L W_K, \quad (3.13)$$

$$V = T_L W_V, \quad (3.14)$$

$$A = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_h}} \right). \quad (3.15)$$

The attended lesion information is added residually to the spatial tokens. If lesion tokens are enabled, the 49 conditioned retinal tokens are concatenated with the four lesion tokens to form a 53-token sequence processed by four Transformer blocks.

3.5.5 Gated Pathology–Global Fusion

The Transformer output is summarized into a retinal representation z_T and a lesion representation z_L . A learned gate controls their relative contribution:

$$g = \sigma(W_g[z_T; z_L] + b_g), \quad (3.16)$$

$$z_{LT} = g \odot z_L + (1 - g) \odot z_T. \quad (3.17)$$

The deepest EfficientNet feature is globally pooled to obtain z_C , while the lesion-burden vector is projected to z_B . The fused representation is

$$z_{full} = [z_C; z_{LT}; z_B], \quad (3.18)$$

and a fusion MLP maps it to the shared 256-dimensional embedding used by the five-class and ordinal heads.

3.6 Multi-Task and Ordinal Learning

3.6.1 Class-Balanced Focal Grading Loss

Training-set counts are converted to effective-number class weights with $\beta = 0.9999$ [92]. The final weights are shown in Table 3.5 together with conventional inverse-frequency weights used only for comparison.

The final grading objective uses the smoother class-balanced weights with focal exponent $\gamma = 2.0$ [91] and label smoothing 0.05.

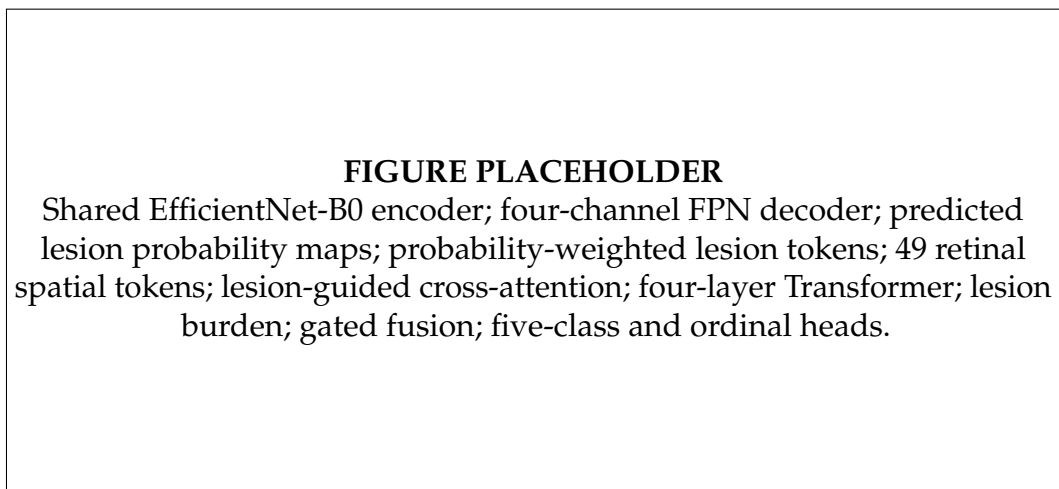


Figure 3.8: Complete lesion-aware CNN-Transformer architecture.

Table 3.5: Training-set imbalance weights.

Grade	Training images	Inverse-frequency	Class-balanced
0	55,162	0.4178	0.7440
1	18,470	1.2479	0.8797
2	24,198	0.9525	0.8133
3	7,936	2.9043	1.3527
4	9,475	2.4325	1.2102

3.6.2 Lesion Segmentation Loss

The segmentation objective combines soft Dice and Focal-Tversky losses [87]:

$$\mathcal{L}_{seg} = 0.5\mathcal{L}_{Dice} + 0.5\mathcal{L}_{FT}. \quad (3.19)$$

For one lesion channel,

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_p P(p)Y(p) + \epsilon}{\sum_p P(p) + \sum_p Y(p) + \epsilon}. \quad (3.20)$$

The Tversky index is

$$TI = \frac{TP + \epsilon}{TP + \alpha FP + \beta FN + \epsilon'}, \quad (3.21)$$

with $\alpha = 0.30$, $\beta = 0.70$, and

$$\mathcal{L}_{FT} = (1 - TI)^{1.33}. \quad (3.22)$$

All four aligned lesion channels are supervised; an empty binary target denotes confirmed absence of the corresponding lesion.

3.6.3 Auxiliary Ordinal Objective

For grade $y \in \{0, 1, 2, 3, 4\}$, the ordinal head predicts four cumulative thresholds:

$$o(y) = [\mathbb{I}(y > 0), \mathbb{I}(y > 1), \mathbb{I}(y > 2), \mathbb{I}(y > 3)]. \quad (3.23)$$

The ordinal loss is binary cross-entropy with logits. The full M4 configuration uses $\lambda_{ord} = 0.20$, while M0–M3 set the coefficient to zero.

3.6.4 Partial-Label Optimization

The two supervision sources are intentionally separated. Grading batches contribute class-balanced focal loss and, in M4, ordinal loss. IDRiD batches contribute lesion-segmentation loss. The general objective can be expressed as

$$\mathcal{L} = m_G(\mathcal{L}_{CBF} + \lambda_{ord}\mathcal{L}_{ord}) + m_S\lambda_{seg}\mathcal{L}_{seg}, \quad (3.24)$$

where m_G and m_S identify available supervision. No pseudo-lesion masks are generated for the combined grading cohort, and Grad-CAM++ is not treated as segmentation ground truth.

3.7 Training Strategy and Baseline Control

3.7.1 Original Grading Optimization

The completed grader uses two-stage transfer learning with AdamW [105]. During warm-up, the EfficientNet backbone is frozen while newly introduced token, Transformer, refinement, projection, and classifier layers are optimized. During full fine-tuning, the backbone is unfrozen with a lower learning rate than the newly introduced layers. Weight decay is 10^{-4} , gradient norm is clipped at 1.0, mixed precision is enabled on CUDA, cosine scheduling is applied, and validation QWK is the checkpoint-selection metric.

3.7.2 Same-Protocol Baseline Control

The selected baseline families span established convolutional and Transformer design principles, including deep residual and densely connected learning, depthwise-separable convolution, modern ConvNets, and multi-axis/hierarchical attention [24, 25, 26, 27, 30, 28].

EfficientNet-B7, Inception-ResNetV2, ConvNeXt, MaxViT, and the proposed hybrid are trained and evaluated using the same locked image-level split and the same experimental protocol. The controlled elements are summarized in Table 3.6.

Table 3.6: Controlled experimental protocol used for all Stage-I grading baselines.

Experimental element	EffNet-B7	Inc-ResNetV2	ConvNeXt	MaxViT	Proposed
Same locked train/validation/test images	Yes	Yes	Yes	Yes	Yes
224×224 input resolution	Yes	Yes	Yes	Yes	Yes
Same retinal preprocessing	Yes	Yes	Yes	Yes	Yes
Same augmentation policy	Yes	Yes	Yes	Yes	Yes
Same class-balanced focal objective	Yes	Yes	Yes	Yes	Yes
Same optimizer/scheduler framework	Yes	Yes	Yes	Yes	Yes
Same validation checkpoint criterion	Yes	Yes	Yes	Yes	Yes
Same evaluation metrics	Yes	Yes	Yes	Yes	Yes

3.7.3 IDRiD Development and Joint Training

The lesion-aware code first performs five-fold development using only the 54 official IDRiD development images. Fold-specific best epochs are selected using validation lesion performance, and the median selected epoch defines final segmentation

initialization on all 54 development images. The 27 official IDRiD test images remain protected until final evaluation.

Joint lesion-aware training uses a grading loader at 512×512 and an IDRiD segmentation loader. The schedule alternates four grading updates for every one segmentation update. For the first three multi-task epochs, decoder parameters are frozen to preserve segmentation initialization. The encoder learning rate is 10^{-5} and non-encoder heads use 10^{-4} , with AdamW, cosine annealing, mixed precision, gradient clipping, and validation-QWK early stopping.

3.8 Evaluation Metrics

The completed grading evaluation reports accuracy, macro-precision, macro-recall, macro-F1, Matthews correlation coefficient where serialized, Cohen’s kappa where available, QWK, macro one-vs-rest ROC-AUC, macro PR-AUC, class-wise sensitivity, specificity, and F1. Ordinal error is defined as

$$e_i = |y_i - \hat{y}_i|, \quad (3.25)$$

with mean absolute grade error, within-one-grade accuracy $P(e_i \leq 1)$, and major-error rate $P(e_i \geq 2)$.

Calibration is evaluated with 15-bin ECE and multiclass Brier score. A positive scalar temperature is fitted on validation logits only and applied to test logits. Predictive uncertainty is summarized by normalized entropy

$$H(p) = -\frac{1}{\log C} \sum_{c=1}^C p_c \log p_c, \quad C = 5. \quad (3.26)$$

The held-out lesion evaluator thresholds sigmoid probabilities at 0.5 and reports Dice, IoU, sensitivity, precision, and pixel-level PR-AUC separately for MA, HE, EX, and SE.

3.9 Experimental and Ablation Protocol

The predefined experiment groups are:

1. dataset structure, provenance, manual harmonization, and class-distribution audit;

2. brightness, contrast, and sharpness characterization;
3. KL split-distribution, t-SNE, Lorenz, and cumulative-distribution analysis;
4. same-protocol five-grade baseline comparison;
5. class-wise performance, confusion matrix, ROC/PR, and ordinal errors;
6. validation-fitted temperature scaling and calibration;
7. CNN/Transformer/spatial branch agreement and progressive prediction analysis;
8. adjacent-grade probability structure and confidence versus error severity;
9. Grad-CAM++ and attention rollout across correct, adjacent-error, and major-error cases;
10. controlled photometric/cropping sensitivity;
11. entropy-error correspondence;
12. IDRiD five-fold segmentation development and held-out lesion evaluation (results pending);
13. lesion-aware grading evaluation and M0–M4 ablation (results pending);
14. lesion-aware calibration evaluation (results pending);
15. RetinaInsight software verification.

The lesion-aware ablation configurations are shown in Table 3.7.

Table 3.7: Predefined M0–M4 lesion-aware ablation configurations.

Model	Seg. supervision	Lesion tokens	Cross-attn.	Ordinal	Purpose
M0	No	No	No	No	512×512 CNN–Transformer baseline
M1	Yes	No	No	No	Isolate auxiliary segmentation
M2	Yes	Yes	No	No	Add pathology-conditioned tokens
M3	Yes	Yes	Yes	No	Add lesion-guided interaction
M4	Yes	Yes	Yes	Yes	Full proposed lesion-aware model

3.10 RetinaInsight Web Application

3.10.1 Purpose and System Architecture

RetinaInsight is a web-based research decision-support prototype built around the completed Stage-I checkpoint. It is designed to preserve inference preprocessing while making five-grade output, probability structure, uncertainty, image-quality indicators, and complementary XAI accessible through a responsive interface. The system has three layers: a browser-based presentation layer, a Flask application/API layer, and a PyTorch inference/XAI layer.

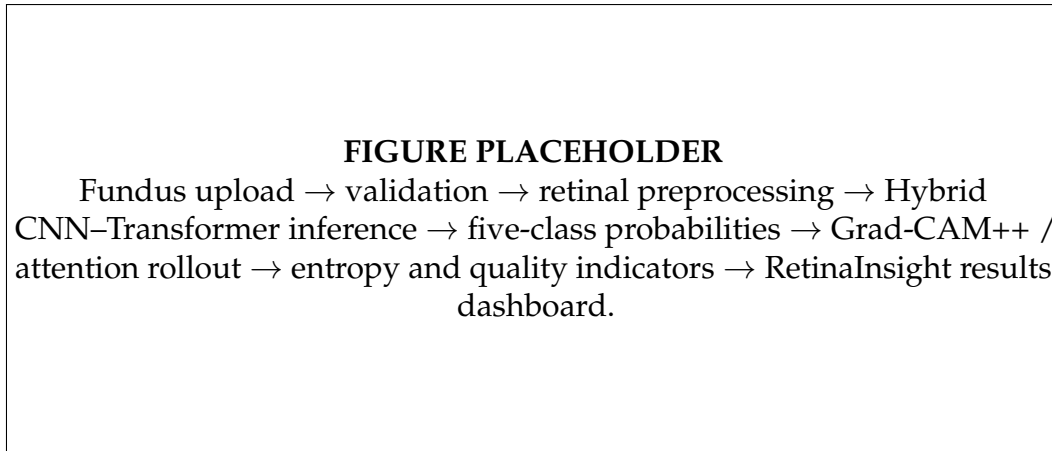


Figure 3.9: RetinaInsight system architecture and end-to-end inference flow.

3.10.2 Model Reconstruction and Inference Consistency

The web application reconstructs the same EfficientNet-B0 + Transformer architecture used during Stage-I evaluation. Key settings include 224×224 input, five output classes, intermediate EfficientNet tap index 3, embedding dimension 256, four Transformer layers, eight attention heads, a 512-unit classifier hidden layer, and 0.50 classifier dropout. The checkpoint state dictionary is loaded with strict key/shape matching.

Inference preprocessing follows the deterministic evaluation transform: RGB conversion, retinal-field cropping, square padding, resize to 224×224 , tensor conversion, and ImageNet normalization. This prevents a silent train/evaluation-to-deployment preprocessing mismatch.

3.10.3 Prediction, Uncertainty, and XAI Output

The application returns the predicted grade, corresponding class name, raw softmax confidence, the full five-class probability vector, normalized entropy, an operational uncertainty category, brightness/contrast/sharpness indicators, the processed image, Grad-CAM++, and Transformer attention rollout. Grad-CAM++ uses the final convolutional feature block, while attention rollout aggregates self-attention matrices across the Transformer layers. Both maps are presented as model-inspection evidence rather than lesion segmentation or causal explanation.

3.10.4 Backend, API, and Input Validation

The Flask backend exposes a main interface, a health endpoint, and a prediction endpoint. File validation checks upload presence, supported extension/type, a 12 MB size limit, decodability, minimum image dimensions, and an upper bound on decompressed image size. The model is loaded once and reused, while a lock serializes prediction/XAI operations that depend on shared gradient hooks and attention state.

3.10.5 Privacy and Clinical-Use Safeguards

Uploaded images are decoded in memory and are not persistently saved to disk by the current application. User-facing errors avoid internal traceback disclosure. The interface explicitly frames output as AI screening/decision support rather than diagnosis, issues a specialist-review warning for high uncertainty, and does not generate treatment recommendations. Image-quality thresholds are operational heuristics rather than clinically validated acceptability criteria.

A key limitation is calibration consistency. The completed experimental analysis fits temperature scaling on validation logits, but the standalone checkpoint does not bundle the fitted temperature scalar. RetinaInsight therefore reports raw softmax probabilities and must not be described as displaying calibrated confidence.

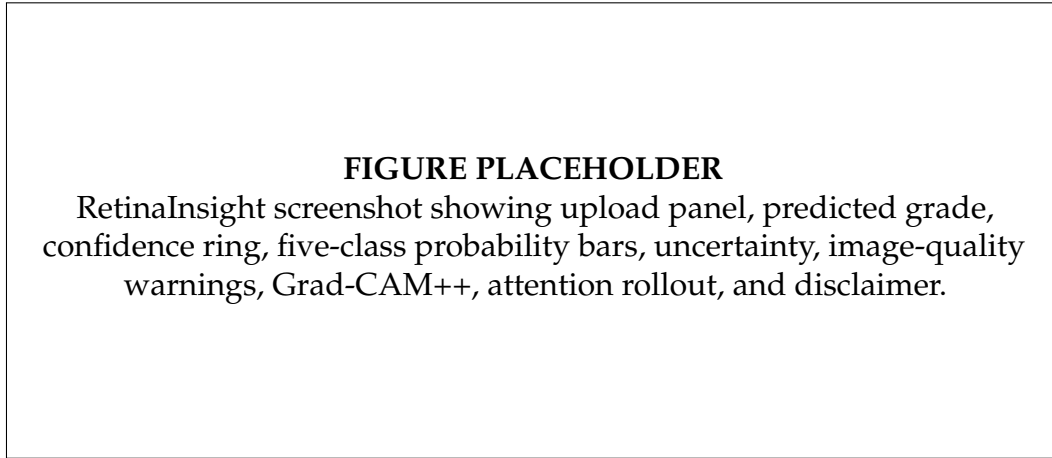


Figure 3.10: RetinaInsight user interface for research-oriented DR inference and explanation.

3.11 Reproducibility, Ethics, Privacy, and Leakage Controls

The grading split is created once and locked. All same-protocol baselines use the same images and optimization framework. Training class weights are calculated from training frequencies only. The grading test split is never used for model selection, and temperature scaling is fitted on validation logits only. Duplicate images are removed during grading-cohort construction, and the lesion-aware integration includes exact and perceptual overlap checks against IDRiD. The IDRiD 54/27 development/test separation is protected. No pseudo-lesion masks are generated for grading-only images.

The study uses existing de-identified retinal image resources and performs no prospective participant recruitment or treatment intervention. The model is evaluated as a research system rather than a medical device. RetinaInsight likewise remains a prototype: it does not maintain patient records, authentication, a clinical audit trail, or regulatory compliance, and uploaded images are not persistently stored by the current implementation. Prospective clinical validation, device/population testing, governance controls, and security hardening would be required before any real medical deployment.

3.12 Chapter Summary

This chapter described the combined APTOS2019–EyePACS–Messidor-2 grading cohort, duplicate removal, manual label harmonization, locked image-level splitting, class and image-quality characterization, the completed Hybrid CNN–Transformer grader, the IDRiD-supervised lesion-aware extension, the M0–M4 experimental design, and the RetinaInsight web prototype. The methodology deliberately separates the completed 224×224 grading experiment from the 512×512 lesion-aware protocol and distinguishes retrospective research performance from web-prototype verification and clinical validation.

Chapter 4

Experimental Results

4.1 Scope of Reported Results

This chapter reports numerical outcomes only for experiments that have been completed under the locked grading protocol and for the implemented RetinaIn-sight software-verification checks. The IDRiD lesion-aware architecture, evaluation metrics, and M0–M4 ablation protocol are fully specified in Chapter 3, but their numerical outcomes are intentionally reserved as TBD until the corresponding experiments are executed. No synthetic or estimated lesion-aware performance values are inserted.

The completed grading experiment evaluates the 14,201-image locked test split after model selection on the 14,227-image validation split. All Stage-I baseline architectures use the same image-level partitions, 224×224 preprocessing, augmentation, class-balanced focal objective, optimizer/scheduler framework, checkpoint criterion, and evaluation metrics.

4.2 Dataset Characterization

4.2.1 Class Distribution and Imbalance

The duplicate-cleaned and manually harmonized APTOS2019–EyePACS–Messidor-2 cohort contains 143,669 images. Grade 0 is the dominant category at 47.99%, while grade 3 is the least represented at 6.90%. The global imbalance ratio is approximately 6.96:1, and the training, validation, and test ratios remain close to this value. The similarity of class concentration across splits indicates that the fixed

partitions preserve the overall imbalance pattern rather than artificially balancing validation or test data.

Table 4.1: Class distribution of the combined grading cohort.

Split	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4	Total
Training	55,162	18,470	24,198	7,936	9,475	115,241
Validation	6,895	1,840	3,024	1,000	1,468	14,227
Test	6,896	1,862	2,999	978	1,466	14,201

4.2.2 Split-Distribution Similarity

KL divergence between train and validation class proportions is approximately 0.0058 and between train and test is approximately 0.0054, with validation–test divergence close to zero. The small divergences show that class proportions are closely aligned across the locked partitions. They do not demonstrate patient-level independence, because the partition was created at image level.

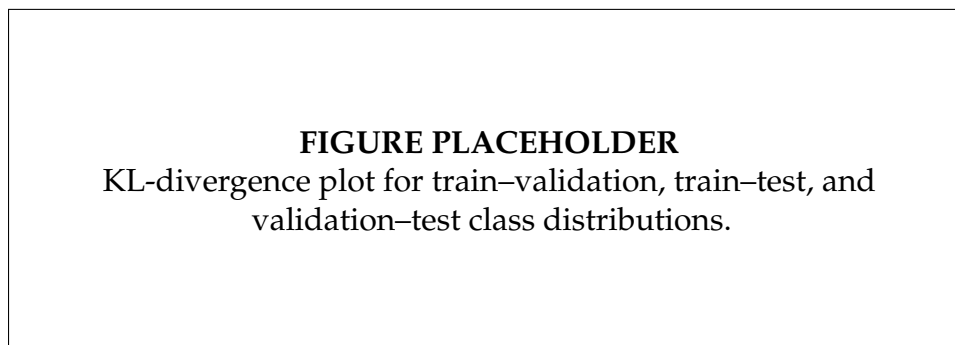


Figure 4.1: Class-proportion similarity across the locked image-level splits.

4.2.3 Brightness, Contrast, and Sharpness

The quality audit reveals substantial acquisition variability rather than a uniform visual distribution. Mean brightness ranges from 101.15 in grade 0 to 110.77 in grade 4, mean contrast ranges from 24.36 to 26.66, and mean sharpness ranges from 43.71 to 65.24 across classes. More important than the means, the underlying distributions contain broad within-class dispersion and outliers. This observation

supports augmentation and sensitivity analysis but does not establish a causal relationship between image quality and prediction error.

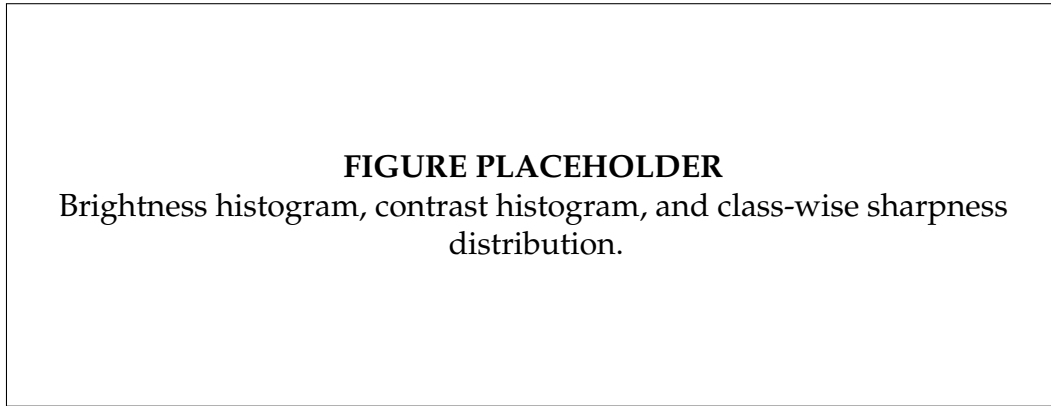


Figure 4.2: Image-quality distributions in the combined grading cohort.

4.2.4 Raw-Feature t-SNE and Class Concentration

In the t-SNE projection of 1,500 randomly sampled 64×64 raw-image vectors, the five grades overlap substantially instead of forming clean severity clusters. The visualization supports the need for learned nonlinear features but is not interpreted as a formal separability test. Lorenz and cumulative-distribution analyses independently confirm that the cohort is concentrated in grades 0–2, with grades 3 and 4 forming a smaller share.

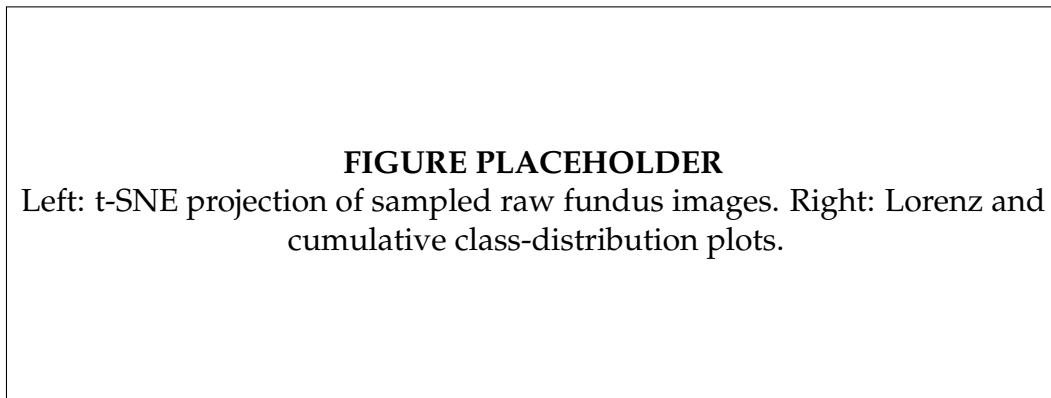


Figure 4.3: Exploratory raw-feature separability and class-concentration analysis.

4.3 Training and Validation Behavior

Training begins with a frozen-backbone warm-up and then proceeds to full fine-tuning. The recorded warm-up completes after 23 epochs, followed by 35 epochs of full fine-tuning. During full fine-tuning, validation macro-F1 rises from 0.5020 at epoch 1 to 0.9620 at epoch 35, while validation QWK increases from 0.6738 to 0.9791. The selected checkpoint is the full-fine-tuning epoch with the highest monitored validation QWK.

Table 4.2: Selected validation checkpoint of the completed hybrid model.

Metric	Value
Validation accuracy	0.9670
Validation macro-F1	0.9620
Validation QWK	0.9791
Validation ECE (15 bins)	0.0302
Validation macro ROC-AUC (OvR)	0.9954
Validation macro PR-AUC	0.9858

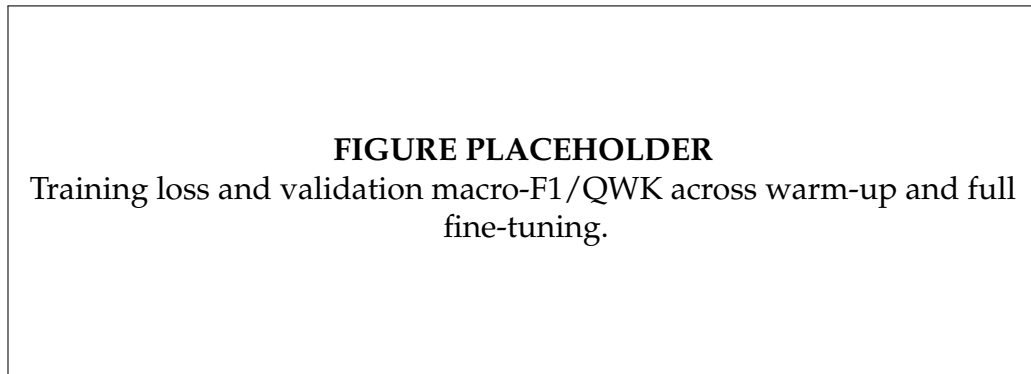


Figure 4.4: Learning behavior of the two-stage Hybrid CNN–Transformer training process.

4.4 Same-Protocol Baseline Comparison

Table 4.3 compares the Hybrid CNN–Transformer with four baseline architectures trained under the same locked Stage-I protocol. Because the data, preprocessing,

augmentation, objective, optimization framework, checkpoint criterion, and evaluation metrics are held constant, this is the appropriate within-study comparison.

Table 4.3: Same-protocol five-class DR grading results on the locked test split.

Model	Accuracy (%)	Macro-P	Macro-R	Macro-F1	QWK
EfficientNet-B7	83.43	0.760	0.460	0.520	–
Inception-ResNetV2	87.31	0.840	0.860	0.850	–
ConvNeXt	91.00	0.920	0.900	0.910	0.9325
MaxViT	93.00	0.950	0.930	0.930	0.9521
Proposed hybrid	96.40	0.9678	0.9600	0.9630	0.9790

Relative to MaxViT, the proposed hybrid improves accuracy by 3.40 percentage points, macro-F1 by 0.033, and QWK by 0.0269. These differences are descriptive rather than inferential because repeated multi-seed training was not performed due to computational limitations.

4.5 Published Five-Class SOTA Context

Table 4.4 places the completed study in the context of recent five-class DR grading literature. Unlike Table 4.3, these studies use different datasets, splits, preprocessing, and averaging conventions. They are therefore presented as contextual SOTA evidence rather than direct experimental competitors.

The published literature also makes clear that headline five-class accuracy depends strongly on the evaluation protocol. Some methods exceed 98% on particular harmonized or internally split cohorts, while other strong architectures report lower performance on DDR, Messidor-2, or external datasets. The same-protocol baseline table is therefore more appropriate for attributing gains within this thesis, while the SOTA table establishes broader methodological context.

4.6 Class-Wise Classification Performance

The class-wise results are reported in Table 4.5. Grade 1 is the weakest class by F1 and sensitivity, while grades 0 and 4 show the strongest separation. The pattern is consistent with the broader literature, where subtle adjacent non-proliferative stages are harder than visually distinct endpoints [53].

Table 4.4: Contextual comparison with representative recent five-class DR grading studies. Values are not directly comparable across heterogeneous protocols.

Study	Year	Dataset(s)	Method	Accuracy	F1	Kappa/QWK	Notes
Gu et al. [12]	2023	DDR / IDRiD	ViT + residual class-specific attention	0.8235 ^a	–	–	Transformer grading
HFF-Net [60]	2024	Public DR grading data	Hierarchical CNN feature fusion	73.77%	–	–	Efficient screening/grading
CRA-Net [13]	2024	DDR / APTOS2019	Lesion/category attention + Transformer guidance	Competitive	Competitive	Reported	Interpretable attention
D-TNet [79]	2025	APTOS2019 / Messidor-2	DenseNet–Transformer fusion	97% / 86%	0.94 / 0.79	0.93 ^b	Domain-shift drop
TOViT [102]	2025	EyePACS / Messidor-2 / APTOS2019	Task-optimized ViT	up to 99%	>0.93	Reported	Resource-constrained deployment
FusionNet [80]	2026	APTOS2019 + Messidor-2; external data	GoogLeNet + ViT fusion	98.85%	97.62% weighted	–	External testing
EffTNet [15]	2026	APTOS2019 + EyePACS + Messidor-2; external data	EfficientNet-B3a + ViT-B16, attention fusion	98.77%	96.99%	0.947	Grad-CAM + external validation

^aPublished average accuracy for the ViT+CSRA ablation configuration on DDR; it is not equivalent to a pooled sample-level accuracy reported under every protocol. ^bKappa reported on APTOS2019. “Competitive” or “Reported” is used where a single comparable aggregate was not consistently available from the cited publication for this table.

Table 4.5: Class-wise performance of the completed hybrid model.

Grade	Precision	Sensitivity	Specificity	F1
0 (No DR)	0.982	0.978	0.989	0.980
1 (Mild)	0.949	0.936	0.991	0.942
2 (Moderate)	0.958	0.955	0.984	0.956
3 (Severe)	0.961	0.948	0.997	0.954
4 (Proliferative)	0.989	0.983	0.998	0.986

Specificity is high for every one-vs-rest class but is influenced by the large number of negative samples. Sensitivity and F1 are therefore more informative for minority-grade behavior. The lower grade-1 sensitivity supports the need for explicit evaluation of adjacent decision boundaries.

4.7 Confusion Matrix and Ordinal Error

The test confusion matrix is dominated by its main diagonal, with most residual errors occurring between neighboring grades. The executed major-error rate is 0.0031, where a major error is defined as $|y - \hat{y}| \geq 2$. Therefore,

$$P(|y - \hat{y}| \leq 1) = 1 - 0.0031 = 0.9969, \quad (4.1)$$

so 99.69% of test predictions fall within one grade of the reference label under the implemented metric definition.

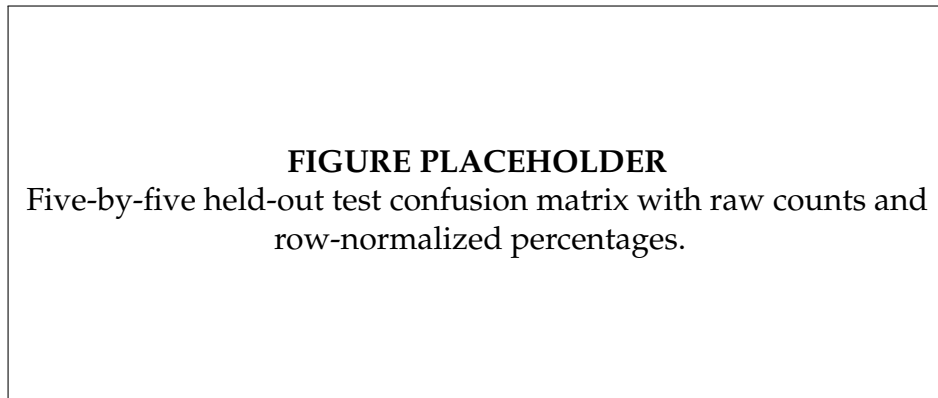


Figure 4.5: Confusion matrix of the completed five-class DR grading model.

The low major-error rate complements QWK by making distant errors directly visible. The exact aggregate grade MAE was not serialized in the completed summary and is therefore not invented.

4.8 ROC and Precision–Recall Analysis

The dedicated probability analysis evaluates one-vs-rest ROC curves for every grade. Grade 2 has the lowest ROC-AUC at 0.9908, while grade 4 reaches 0.9989. Macro-average ROC-AUC is 0.9954, micro-average ROC-AUC is 0.9961, and support-weighted OvR ROC-AUC is 0.9957.

Table 4.6: Class-wise and aggregate ROC-AUC values.

Analysis	ROC-AUC
Grade 0	0.9982
Grade 1	0.9941
Grade 2	0.9908
Grade 3	0.9950
Grade 4	0.9989
Macro average	0.9954
Micro average	0.9961
Weighted OvR	0.9957

The macro PR-AUC is 0.9858. PR-AUC, class-wise sensitivity, macro-F1, and calibration are retained alongside ROC-AUC because ROC-AUC can appear optimistic in imbalanced settings.

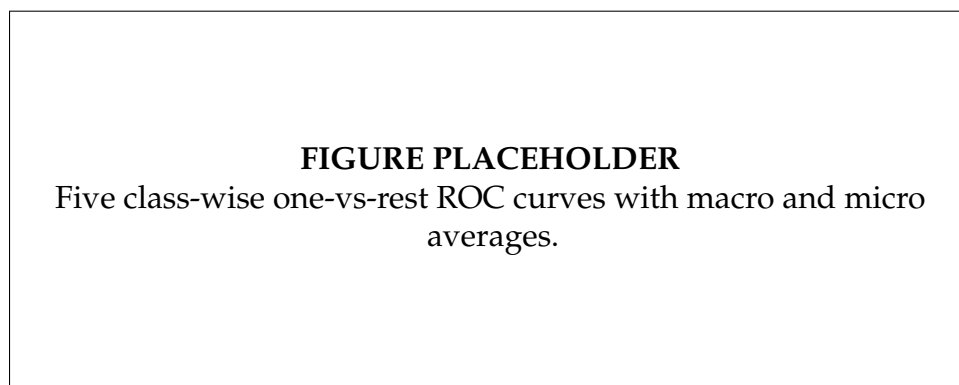


Figure 4.6: One-vs-rest ROC curves for five DR severity grades.

4.9 Calibration

The uncalibrated classifier achieves ECE 0.0610 and multiclass Brier score 0.0660. A single positive temperature is fitted using validation logits only and then applied to the locked test logits. Temperature scaling leaves class ordering unchanged and therefore preserves accuracy, macro-F1, QWK, ROC-AUC, PR-AUC, and major-error rate, while reducing ECE to 0.0240 and Brier score to 0.0580.

The result demonstrates improved retrospective probability calibration rather than improved categorical discrimination. No clinical referral threshold is validated

Table 4.7: Test performance before and after validation-fitted temperature scaling.

Metric	Uncalibrated	Temperature-scaled
Accuracy	0.9640	0.9640
Macro-Precision	0.9678	0.9678
Macro-Recall	0.9600	0.9600
Macro-F1	0.9630	0.9630
QWK	0.9790	0.9790
Macro ROC-AUC	0.9954	0.9954
Macro PR-AUC	0.9858	0.9858
Major-error rate	0.0031	0.0031
ECE	0.0610	0.0240
Brier score	0.0660	0.0580

in this study.

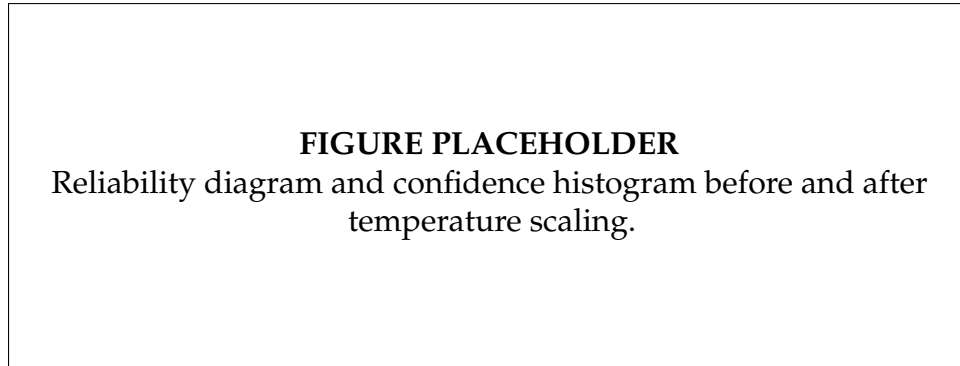


Figure 4.7: Calibration behavior before and after temperature scaling.

4.10 Local–Global Agreement and Progressive Prediction Refinement

Analytical probe heads attached to the CNN-global, Transformer-global, and spatial-refinement representations provide branch-level predictions without changing the trained final model. Representative cases show that the branches can disagree and that the fused representation can shift probability toward the reference class. These figures are interpretive and are not treated as independently trained component ablations.

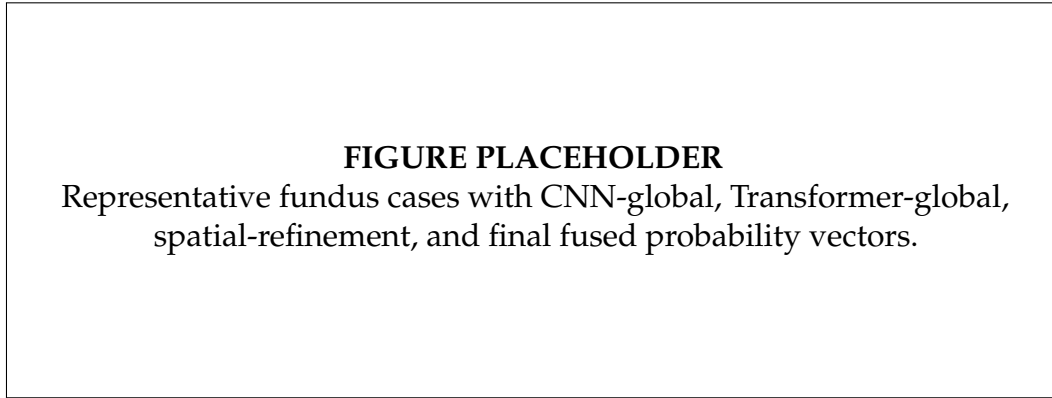


Figure 4.8: Local-global branch agreement and representative disagreement resolution.

A second visualization follows the probability progression from CNN representation to Transformer representation, spatial refinement, and final fusion. This analysis helps inspect how probability mass moves among neighboring grades as richer representations are incorporated.

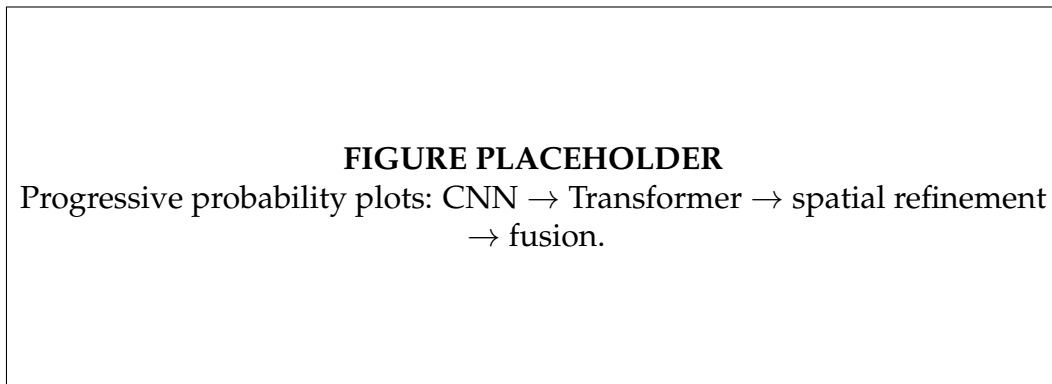


Figure 4.9: Progressive prediction refinement across internal representations.

4.11 Adjacent-Grade Decision Analysis

Representative cases are selected around the four adjacent boundaries 0–1, 1–2, 2–3, and 3–4. The Mild–Moderate boundary is particularly important because grade 1 has the lowest F1 and sensitivity, while grade 2 has the lowest one-vs-rest ROC-AUC among the five classes. Probability plots are interpreted as model

uncertainty near ordinal decision boundaries rather than as calibrated clinical risk before temperature scaling.

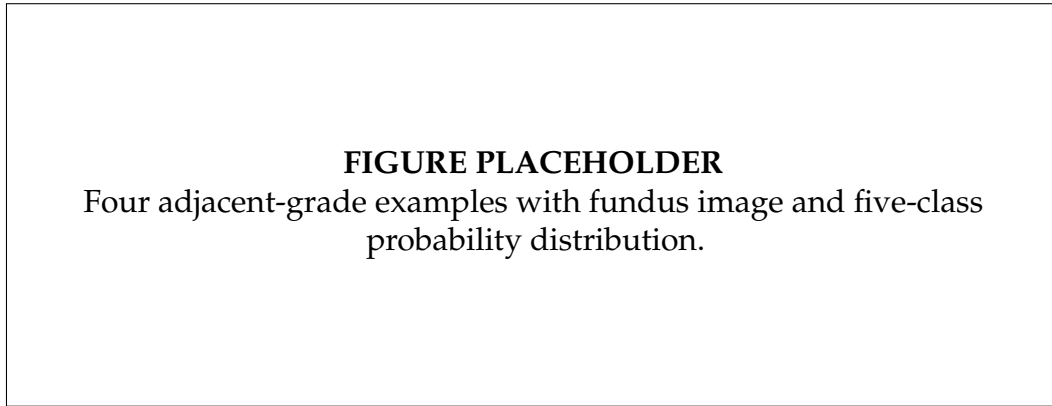


Figure 4.10: Representative adjacent-grade decision boundaries.

4.12 Class-Wise Probability Structure

A 5×5 matrix of mean predicted probabilities conditioned on the true grade complements the confusion matrix by preserving non-argmax probability mass. The expected pattern is strong diagonal concentration with secondary probability primarily in neighboring classes. The final exported figure should be inserted directly from the probability-analysis output rather than manually approximating cell values.

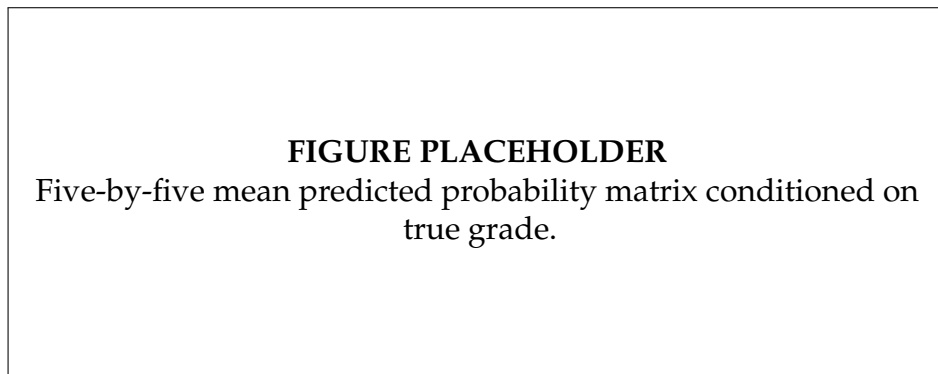


Figure 4.11: Mean class-probability structure for each reference DR grade.

4.13 Confidence and Error Severity

Predictions are grouped as correct ($e = 0$), adjacent errors ($e = 1$), and major errors ($e \geq 2$). Confidence distributions overlap and include some high-confidence errors, showing that maximum softmax probability alone is not an error detector. Median confidence nevertheless decreases with increasing ordinal error severity at the population level, motivating the complementary calibration and entropy analyses.

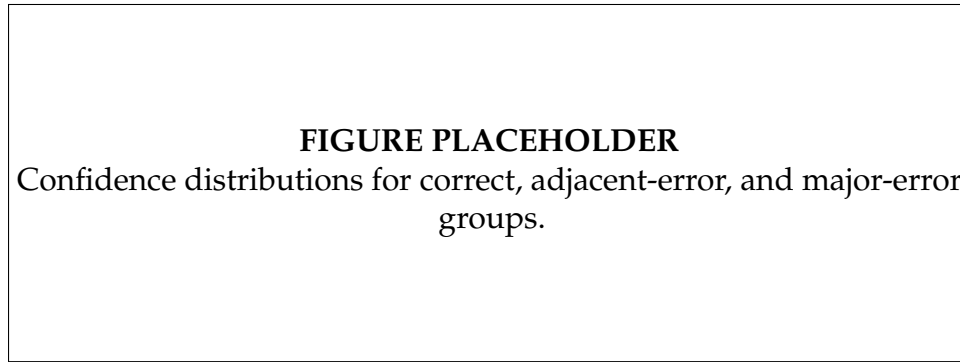


Figure 4.12: Prediction confidence as a function of ordinal error severity.

4.14 Explainability Across Correct and Error Cases

Representative correct, adjacent-error, and major-error cases are visualized with the original fundus image, five-class probabilities, Grad-CAM++, and Transformer attention rollout. Grad-CAM++ tends to emphasize localized regions, while attention rollout provides broader token-level context. The maps are interpreted conservatively as model-sensitivity evidence rather than lesion masks or causal proof. The possibility of visually plausible explanations for incorrect predictions is one reason the second-stage framework uses explicit IDRiD lesion supervision.

4.15 Perturbation Sensitivity

A representative correctly classified image is evaluated after reduced brightness, reduced contrast, mild Gaussian blur, reduced color saturation, and mild cropping. Brightness reduction, blur, and color variation produce relatively smaller confidence changes, whereas low contrast and cropping produce larger confidence reductions

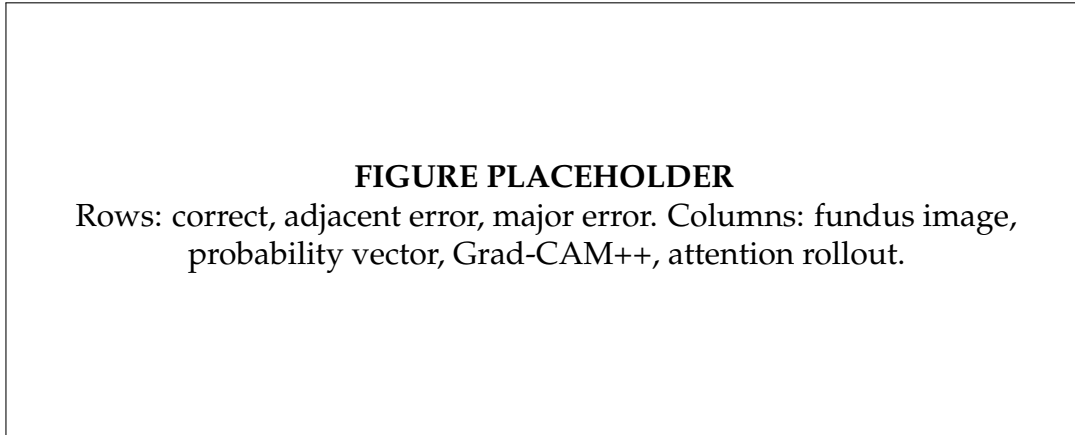


Figure 4.13: Post-hoc explanations across correct and ordinally incorrect predictions.

and redistribute probability toward neighboring grades. Because the analysis is based on controlled transformations rather than an external domain, it is described as sensitivity analysis rather than proof of deployment robustness.

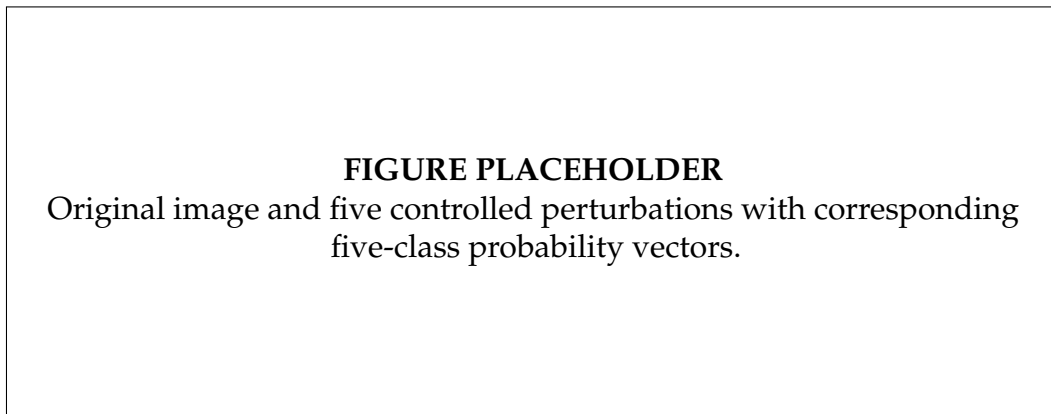


Figure 4.14: Prediction sensitivity to controlled image perturbations.

4.16 Entropy–Error Correspondence

All 14,201 test cases are divided into five approximately equal-sized groups by normalized predictive entropy. Table 4.8 shows a monotonic relationship between uncertainty and error. Accuracy decreases from 0.9961 in the lowest-entropy group to 0.9122 in the highest-entropy group; mean absolute class error increases from 0.0039 to 0.1233; and major-error rate increases from 0 to 0.0137.

Table 4.8: Performance across normalized predictive-entropy quintiles.

Entropy group	Mean entropy	Accuracy	Mean abs. error	Major-error rate	<i>n</i>
Lowest	0.0755	0.9961	0.0039	0.0000	2,846
Low	0.1704	0.9902	0.0098	0.0000	2,845
Medium	0.2811	0.9789	0.0221	0.0004	2,845
High	0.4192	0.9582	0.0443	0.0014	2,845
Highest	0.6407	0.9122	0.1233	0.0137	2,846

Entropy is therefore a useful continuous indicator of population-level difficulty, but it is not a binary error detector because the highest-entropy group still retains accuracy above 0.91.

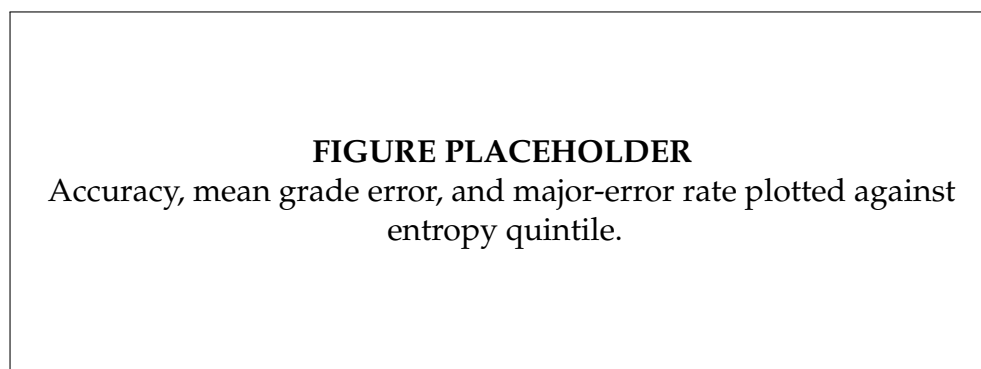


Figure 4.15: Relationship between predictive entropy and grading error severity.

4.17 RetinaInsight Implementation Verification

RetinaInsight translates the completed 224×224 grading checkpoint into a responsive Flask/PyTorch web workflow. The implementation verification checks syntax, strict model reconstruction, API readiness, end-to-end inference, probability normalization, XAI generation, and CPU/GPU execution. Table 4.9 summarizes the completed engineering checks.

The current application exposes raw softmax probabilities, not the validation-fitted temperature-scaled probabilities used in the experimental calibration analysis. It also provides grading rather than lesion segmentation, has no patient-record database or authentication layer, and is therefore presented as a research decision-support prototype.

Table 4.9: RetinaInsight software implementation verification. These checks establish software functionality, not clinical validity.

Verification item	Status
Python application/model syntax	Pass
JavaScript syntax	Pass
Strict checkpoint state-dict loading	Pass
Model metadata: 7×7 tokens and five outputs	Pass
Health endpoint	HTTP 200
End-to-end upload $\rightarrow$ prediction/XAI response	HTTP 200
Five-class probability normalization	Pass
Original/Grad-CAM++/attention image payload generation	Pass
CPU execution	Pass
CUDA execution when available	Pass

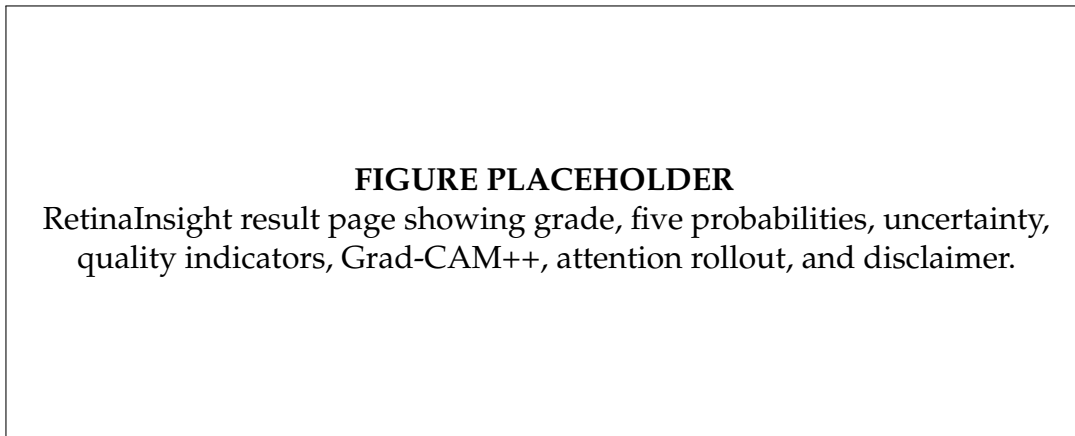


Figure 4.16: RetinaInsight research decision-support interface.

4.18 Integrated Discussion

The completed experiments provide evidence for four main conclusions about Stage I. First, the Hybrid CNN–Transformer achieves strong five-grade discrimination on a large, imbalanced combined cohort, with 96.40% accuracy, 0.9630 macro-F1, and 0.9790 QWK. Because EfficientNet-B7, Inception-ResNetV2, ConvNeXt, MaxViT, and the proposed model use the same locked protocol, the observed within-study differences are more informative than cross-paper SOTA comparisons. The proposed model’s advantage over MaxViT is therefore interpreted as a controlled descriptive improvement, while the lack of repeated seeds prevents inferential significance claims.

Second, the class-wise and ordinal results show that performance is not uniform across severity. Mild DR remains the weakest class, consistent with recent meta-analytic evidence that adjacent non-proliferative grades are difficult to distinguish [53]. The major-error rate of 0.0031 indicates that distant disagreements are rare even though adjacent ambiguity remains.

Third, reliability cannot be inferred from accuracy alone. Temperature scaling substantially improves ECE and Brier score without altering predictions, while entropy quintiles show a monotonic relationship between uncertainty and both error frequency and error magnitude. This supports the use of calibration and uncertainty as complementary evaluation dimensions rather than treating raw maximum probability as sufficient.

Fourth, local–global probes and XAI provide useful model-inspection evidence but do not establish explicit pathological grounding. Grad-CAM++ and attention rollout can highlight plausible retinal regions while remaining post-hoc and potentially misleading on individual errors. This motivates the lesion-aware extension, whose novelty is not simply an auxiliary FPN decoder but the direct transformation of supervised lesion maps into pathology-conditioned tokens that modulate retinal Transformer reasoning through cross-attention and gated fusion.

The published SOTA table also illustrates why direct cross-paper ranking should be avoided. Recent DR domain-generalization studies specifically document the difficulty of preserving performance and calibration across unseen fundus domains [70, 71]. Recent models report strong five-class accuracy on APTOS2019 or harmonized cohorts but can show lower performance on DDR, Messidor-2, or

external data [79, 15]. The present thesis therefore separates contextual literature performance from the controlled baseline comparison.

Finally, RetinaInsight demonstrates that the completed model can be reconstructed outside the training environment and delivered through an end-to-end inference/XAI workflow with explicit privacy and clinical-use safeguards. The software verification does not establish clinical effectiveness, and the current use of raw rather than calibrated softmax probabilities is stated transparently. Its contribution is translational reproducibility rather than medical-device validation.

4.19 Reserved Lesion-Aware Tables (Results Pending)

The following tables were defined before observing lesion-aware numerical outcomes. They will be populated only from the locked IDRiD/512×512 workflow.

Table 4.10: Reserved IDRiD five-fold segmentation-development results.

Fold	MA Dice	HE Dice	EX Dice	SE Dice	Mean Dice	Best epoch
1	TBD	TBD	TBD	TBD	TBD	TBD
2	TBD	TBD	TBD	TBD	TBD	TBD
3	TBD	TBD	TBD	TBD	TBD	TBD
4	TBD	TBD	TBD	TBD	TBD	TBD
5	TBD	TBD	TBD	TBD	TBD	TBD
Mean	TBD	TBD	TBD	TBD	TBD	–

Table 4.11: Reserved held-out IDRiD lesion-segmentation evaluation.

Lesion	Dice	IoU	Sensitivity	Precision	PR-AUC
MA	TBD	TBD	TBD	TBD	TBD
HE	TBD	TBD	TBD	TBD	TBD
EX	TBD	TBD	TBD	TBD	TBD
SE	TBD	TBD	TBD	TBD	TBD

No interpretation is provided for the reserved tables at this stage.

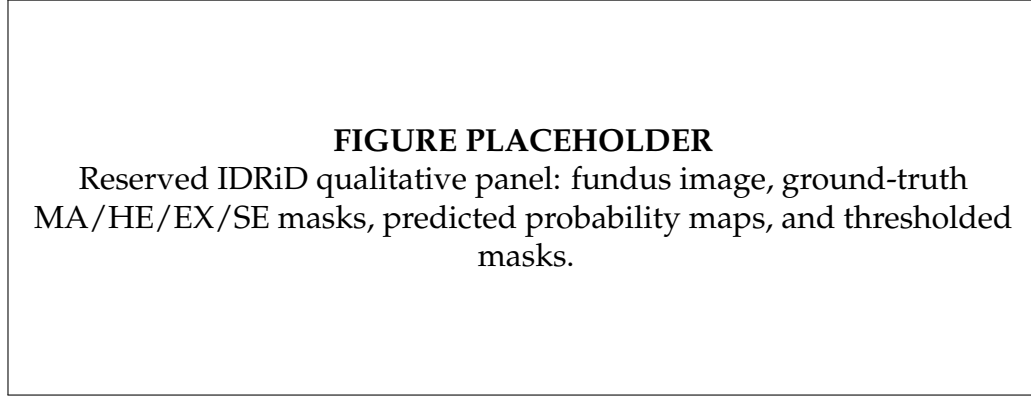


Figure 4.17: Reserved qualitative lesion-segmentation results for the held-out IDRiD test set.

Table 4.12: Reserved M0–M4 lesion-aware ablation results.

Model	Seg.	Tokens	Cross-attn.	Ordinal	Accuracy	Macro-F1	QWK	Grade MAE
M0	No	No	No	No	TBD	TBD	TBD	TBD
M1	Yes	No	No	No	TBD	TBD	TBD	TBD
M2	Yes	Yes	No	No	TBD	TBD	TBD	TBD
M3	Yes	Yes	Yes	No	TBD	TBD	TBD	TBD
M4	Yes	Yes	Yes	Yes	TBD	TBD	TBD	TBD

Table 4.13: Reserved comparison between the completed 224×224 hybrid grader and full 512×512 lesion-aware M4 configuration. This is not a one-factor ablation because image resolution and training protocol differ.

Model	Accuracy	Macro-P	Macro-R	Macro-F1	MCC	QWK	Grade MAE	Within ± 1
Original hybrid (224×224)	0.9640	0.9678	0.9600	0.9630	–	0.9790	–	0.9969
Lesion-aware M4 (512×512)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 4.14: Reserved class-wise grading performance for lesion-aware M4.

Grade	Precision	Sensitivity	Specificity	F1
0	TBD	TBD	TBD	TBD
1	TBD	TBD	TBD	TBD
2	TBD	TBD	TBD	TBD
3	TBD	TBD	TBD	TBD
4	TBD	TBD	TBD	TBD

Table 4.15: Reserved lesion-aware calibration results.

Metric	Before calibration	Temperature-scaled
Accuracy	TBD	TBD
Macro-F1	TBD	TBD
QWK	TBD	TBD
Macro ROC-AUC	TBD	TBD
Macro PR-AUC	TBD	TBD
ECE	TBD	TBD
Brier score	TBD	TBD

Table 4.16: Reserved ordinal-error comparison for the lesion-aware experiment.

Model	Exact-grade accuracy	Within ± 1	Major-error rate	Grade MAE
M0	TBD	TBD	TBD	TBD
M4	TBD	TBD	TBD	TBD

4.20 Chapter Summary

The completed Hybrid CNN–Transformer achieves 96.40% test accuracy, 0.9630 macro-F1, and 0.9790 QWK on the locked image-level grading test set. Same-protocol baseline control supports a fair within-study comparison, while the literature SOTA table provides separate contextual evidence. Temperature scaling improves retrospective calibration without changing predictions, and predictive entropy increases systematically with ordinal error severity. Explainability and branch-level analyses support complementary local and global representations while preserving the distinction between post-hoc attribution and explicit lesion supervision. RetinaInsight successfully reconstructs the completed grading pipeline as a web-based research prototype. Numerical outcomes for the IDRiD lesion-aware extension remain intentionally pending.

Chapter 5

Conclusion

5.1 Summary of the Study

This work examined automated five-grade diabetic retinopathy (DR) assessment from color fundus photographs through a staged research framework. The completed grading stage combines APTOS2019, EyePACS, and Messidor-2 into a duplicate-cleaned, manually harmonized five-grade cohort of 143,669 images. A single image-level train/validation/test split was created once and then kept fixed for the proposed model and every grading baseline. The final Hybrid CNN–Transformer uses EfficientNet-B0 to extract local retinal features, converts an intermediate feature map into a compact set of spatial tokens, models long-range context with a four-block Transformer, refines the spatial tokens convolutionally, and fuses Transformer, spatial, and global CNN representations for five-class prediction.

Evaluation of the completed grader extends well beyond accuracy. The reported analysis covers macro-averaged performance, class-wise sensitivity and specificity, QWK, major ordinal errors, ROC and PR behavior, validation-fitted calibration, branch-level prediction patterns, confidence, predictive entropy, post-hoc explainability, and controlled perturbation sensitivity. A separate literature SOTA analysis provides contextual comparison, while EfficientNet-B7, Inception-ResNetV2, ConvNeXt, and MaxViT serve as controlled same-protocol experimental baselines.

The second methodological stage moves from implicit lesion sensitivity toward explicit pathology-conditioned representation learning. IDRiD provides four aligned binary channels for microaneurysms, hemorrhages, hard exudates, and soft exudates. An FPN decoder predicts lesion probability maps, which are then

converted into probability-weighted lesion tokens and lesion-burden features. Retinal spatial tokens query these pathology-specific tokens through lesion-guided cross-attention, after which gated fusion combines lesion information with broader retinal context before five-class and ordinal prediction. The full M0–M4 protocol is predefined, while the corresponding numerical lesion-aware outcomes remain pending.

The third contribution, *RetinaInsight*, reconstructs the completed grading architecture in a Flask/PyTorch web application. It presents five-grade predictions, all-class probabilities, Grad-CAM++, Transformer attention rollout, normalized entropy, operational uncertainty, and basic image-quality indicators. In the current implementation, uploaded images are kept in memory rather than stored persistently.

5.2 Principal Findings of Stage I

On the locked 14,201-image test set, the completed Hybrid CNN–Transformer achieves 96.40% accuracy, 0.9600 macro-recall, 0.9630 macro-F1, and 0.9790 QWK. Macro-precision is 0.9678, macro ROC-AUC is 0.9954, and macro PR-AUC is 0.9858. Under the same locked protocol, the proposed model exceeds the reported EfficientNet-B7, Inception-ResNetV2, ConvNeXt, and MaxViT baselines. Because repeated multi-seed training was not performed due to computational limitations, these differences are interpreted descriptively rather than as inferentially significant.

The class-wise results show that Mild DR is the most difficult category, with $F1=0.942$ and $sensitivity=0.936$. No DR and Proliferative DR are more clearly separated, with F1 scores of 0.980 and 0.986, respectively. This pattern is consistent with the broader five-class DR literature, where adjacent non-proliferative stages remain more difficult than visually distinct endpoint classes [68, 78, 12].

Ordinal analysis adds another view of reliability. With major error defined as $|y - \hat{y}| \geq 2$, the major-error rate is 0.0031, meaning that 99.69% of test predictions lie within one grade of the reference label. Validation-fitted temperature scaling reduces ECE from 0.0610 to 0.0240 and multiclass Brier score from 0.0660 to 0.0580 without changing the predicted classes. Predictive entropy follows a similarly ordered pattern: accuracy falls from 99.61% in the lowest-entropy quintile to 91.22% in the highest-entropy quintile, while major-error rate rises from 0 to approximately

1.37%. These findings support calibration and uncertainty as complementary reliability measures rather than substitutes for one another or for raw maximum softmax confidence.

The local–global and XAI analyses are also informative, although their interpretation must remain cautious. CNN-derived features and Transformer attention provide complementary views of the retinal image, but Grad-CAM++ and attention rollout are post-hoc methods and can still appear visually plausible when the underlying prediction is wrong. That limitation provides a direct motivation for the explicitly lesion-supervised second stage.

5.3 Methodological Contribution of the Lesion-Aware Extension

The lesion-aware design is more than an auxiliary segmentation branch attached to the grader. Its main methodological contribution lies in how explicit lesion probability maps are carried into the grading representation. For each lesion channel, probability-weighted pooling generates a learned token from an intermediate retinal feature map. The four lesion tokens then act as keys and values in cross-attention with retinal spatial tokens. A learned gate balances lesion-specific information against the broader retinal Transformer representation, while lesion-burden features provide an additional differentiable summary. The resulting shared representation feeds both a five-class grading head and an auxiliary ordinal head.

This architecture leads to a specific experimental hypothesis: explicit MA, HE, EX, and SE supervision may provide pathology-grounded information that complements the local–global representation learned from image-level grading labels. The M0–M4 ablation sequence isolates auxiliary segmentation, lesion tokens, lesion-guided cross-attention, and ordinal learning under the same 512×512 protocol. Numerical claims are intentionally deferred until those experiments are executed.

5.4 RetinaInsight as a Translational Research Contribution

RetinaInsight shows how the completed grading model can be reconstructed outside the original development environment and exposed through a reproducible

browser-to-model workflow. The application preserves deterministic 224×224 evaluation preprocessing, uses strict checkpoint loading, returns grade 0–4 probabilities, generates Grad-CAM++ and attention rollout, reports normalized entropy and image-quality indicators, and performs input validation with privacy-conscious in-memory image handling.

Software verification covers Python and JavaScript syntax, strict model loading, health and prediction endpoints, probability normalization, XAI generation, and CPU/CUDA execution. These checks establish implementation functionality, not clinical validity. The standalone prototype currently reports raw softmax probabilities because the fitted temperature scalar is not bundled with the checkpoint. It also does not provide lesion segmentation, patient records, authentication, database storage, or a clinical audit trail. For these reasons, RetinaInsight is presented as a research decision-support prototype rather than a diagnostic medical device.

5.5 Limitations

The completed findings and the planned lesion-aware extension should be interpreted in light of several limitations. First, the combined grading cohort is split at image level. Duplicate images were removed before the final cohort was established, but patient or eye identity was not used to enforce subject-disjoint separation. Reference-standard and distribution-shift studies show why independence, annotation quality, and external-domain evaluation materially affect the interpretation of reported performance [64, 99, 100]. The reported results therefore should not be read as evidence of patient-level independence.

Second, each grading architecture was evaluated through one locked training run. This constraint reflects the available computational resources, but repeated multi-seed training would provide a stronger estimate of optimization variability. The baseline differences are therefore reported descriptively, without statistical significance claims.

Third, APTOS2019, EyePACS, and Messidor-2 contribute heterogeneous acquisition conditions. The image-quality audit and fixed split characterize this variation, but the completed grading experiment does not isolate source-specific generalization or provide prospective camera-domain validation.

Fourth, the original grading pipeline operates at 224×224 , whereas the lesion-aware extension operates at 512×512 to preserve smaller lesion structures. A direct comparison between the completed original model and M4 therefore changes both architecture and resolution. The M0–M4 sequence is the appropriate within-protocol experiment for attributing effects to lesion-aware components.

Fifth, the IDRiD pixel-annotated segmentation cohort is small relative to the grading dataset. The official 54-image development and 27-image test separation protects the held-out lesion evaluation, but the limited annotation volume may still constrain lesion generalization.

Finally, the work remains retrospective. Responsible medical-AI guidance emphasizes staged clinical evaluation, explicit workflow boundaries, and evidence beyond retrospective discrimination before deployment claims are justified [51, 52]. Calibration, uncertainty, lesion segmentation, XAI, and a functioning web interface improve methodological transparency, but they do not establish clinical safety, workflow effectiveness, or regulatory readiness. RetinaInsight also lacks authentication, audit logging, patient-record integration, deployment-grade model monitoring, and the fitted calibration scalar in its current standalone checkpoint artifact.

5.6 Future Work

The most immediate next step is to complete the predefined IDRiD segmentation, lesion-aware grading, M0–M4 ablation, and lesion-aware calibration experiments without changing the locked evaluation definitions. Repeated multi-seed runs can be added later if computational resources permit.

External lesion datasets would provide a stronger test of MA, HE, EX, and SE localization across devices, populations, and annotation protocols. Where patient identifiers are available, patient-level separation should be enforced. Lesion-specific leave-one-out ablations, lesion-burden versus grade analysis, and comparison of segmentation quality across correct, adjacent-error, and major-error grading groups would help determine whether explicit lesion localization is meaningfully connected to the final severity decision.

Reliability analysis could also move beyond single uncertainty indicators. Entropy, calibrated confidence, branch disagreement, and lesion-prediction uncertainty could be combined into referral-coverage and residual-risk analyses. For

RetinaInsight, the fitted temperature parameter should be versioned with the model artifact so that the web interface can report calibrated probabilities. After completion of the Stage-II experiments, a validated lesion-aware M4 checkpoint could also be integrated to provide lesion-map visualization.

Further software development should address authentication, authorization, secure transport, rate limiting, structured audit logging, model/version tracking, drift monitoring, containerized deployment, automated tests, rollback support, and clinician-feedback workflows. Prospective clinical evaluation remains necessary before any deployment-oriented claim can be supported.

5.7 Final Conclusion

Overall, the work traces a progression from large-scale five-grade DR classification toward reliability-aware, pathology-conditioned, and translational retinal AI. The completed Hybrid CNN–Transformer demonstrates strong discrimination and ordinal agreement on a duplicate-cleaned, manually harmonized APTOS2019–EyePACS–Messidor-2 cohort under a controlled same-protocol baseline design. Calibration and uncertainty analyses show why accuracy alone is insufficient for assessing reliability, while the local–global and post-hoc XAI analyses reveal both complementary representations and the limits of attribution-based pathology claims.

The IDRiD-supervised extension addresses that limitation by transforming explicit lesion probability maps into lesion-conditioned tokens that modulate retinal Transformer reasoning through cross-attention and gated fusion. Its evaluation protocol is predefined, and no numerical benefit is claimed before the experiments are run. RetinaInsight translates the completed grading model into an accessible research prototype while preserving inference consistency and explicit clinical-use boundaries. Together, these components provide a coherent framework for DR grading, reliability analysis, explicit lesion-aware representation learning, and reproducible research translation.

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Thesis for the degree of

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by

**Student Name
Student ID**



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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